EELU

The Egyptian E-Learning University

Faculty of Computers & Information Technology — Fayoum Branch

Graduation Project

AI-Driven Secure Enterprise Infrastructure and Monitoring

Network Infrastructure Design and Security Report

Zero-Trust Network Infrastructure & Security Implementation Report

Track	Network & Network Security
Environment	PNETLab
Architecture	Zero-Trust · Defense-in-Depth

Defense-in-Depth | Zero-Trust Architecture | PNETLab

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Chapter 1: Introduction

1.1 Overview

This report documents the engineering work carried out to design and secure the enterprise network infrastructure as part of the graduation project, based on the Zero-Trust Architecture model, which rests on the principle of never trusting any device, user, or data path before it has been verified and subjected to security policies.

The work was divided into consecutive strategic phases that deliver layered protection (Defense-in-Depth), with the engineering rationale and security importance of every technology documented, so that each configuration can be explained, reviewed, and reproduced.

This work also lays the groundwork for the security monitoring and analysis phase using the Wazuh SIEM platform and AI models; a stable, isolated infrastructure ensures that logs reach the analysis platform without interruption and that the source of any security alert is identified accurately.

1.2 Objectives

- Build a stable, highly available network environment that guarantees a continuous flow of data and logs.
- Divide the network into 7 isolated security zones to reduce the attack surface and prevent lateral movement.

- Secure user-facing access ports as the first line of defense against internal threats.
- Harden the devices administratively (device hardening) and apply the principle of least privilege.
- Delegate inter-VLAN routing to the FortiGate so that every exchange of data between departments is subject to security inspection.
- Deploy a honeypot in an isolated zone to detect reconnaissance and attacks early without putting the real assets at risk.
- Prepare the infrastructure for integration with Wazuh SIEM and the artificial intelligence team.

1.3 Architectural Principles

The network design is based on four main principles, summarized in the following table together with how each is applied in the project:

Principle	Concept	Application in the Project
Zero-Trust	No device or path is trusted by default; every communication is verified and inspected.	Disabling inter-VLAN routing on Core_SW and forcing all inter-department traffic through the FortiGate.
Defense-in-Depth	Multiple protection layers, so that the failure of one layer does not expose the network.	Layer 2 protection, Management Plane hardening, and Layer 3 inspection at the Firewall.
Least Privilege	Granting only the minimum privileges required to perform a task.	Restricting administrative access to the FortiGate to the IT_SecOps network and allowing only PING from all other networks.
High Availability	Service continuity when a component or path fails.	Link aggregation with EtherChannel to eliminate any Single Point of Failure.

1.4 Implementation Environment

The project was implemented entirely within the PNETLab simulation environment. The network consists of the following components:

Component	Technology	Role
Core_SW	Cisco IOL	The backbone of the network; acts as the Root Bridge and aggregates all links.
Server_SW	Cisco IOL	Connects the server zone (VLAN 40 and VLAN 60), with VLAN 7 for device management.
Access_SW_1	Cisco IOL	User-facing ports on the first floor.

Component	Technology	Role
Access_SW_2	Cisco IOL	User-facing ports on the second floor.
FortiGate	QEMU Virtual Appliance	Firewall, central Layer 3 router, and DHCP server.
Windows Server	Active Directory	Active Directory services for the AegisEud.local domain in VLAN 40.
Wazuh Manager	Wazuh SIEM	Monitoring and log-analysis platform in VLAN 40.
Honeypot	Decoy Server	Isolated decoy server in VLAN 60 for analyzing attacker behavior.

1.5 Network Topology

The following figure shows the overall network topology: the FortiGate as the central gateway, Core_SW as the backbone, Server_SW for the server zone, and two Access switches distributed across two floors.

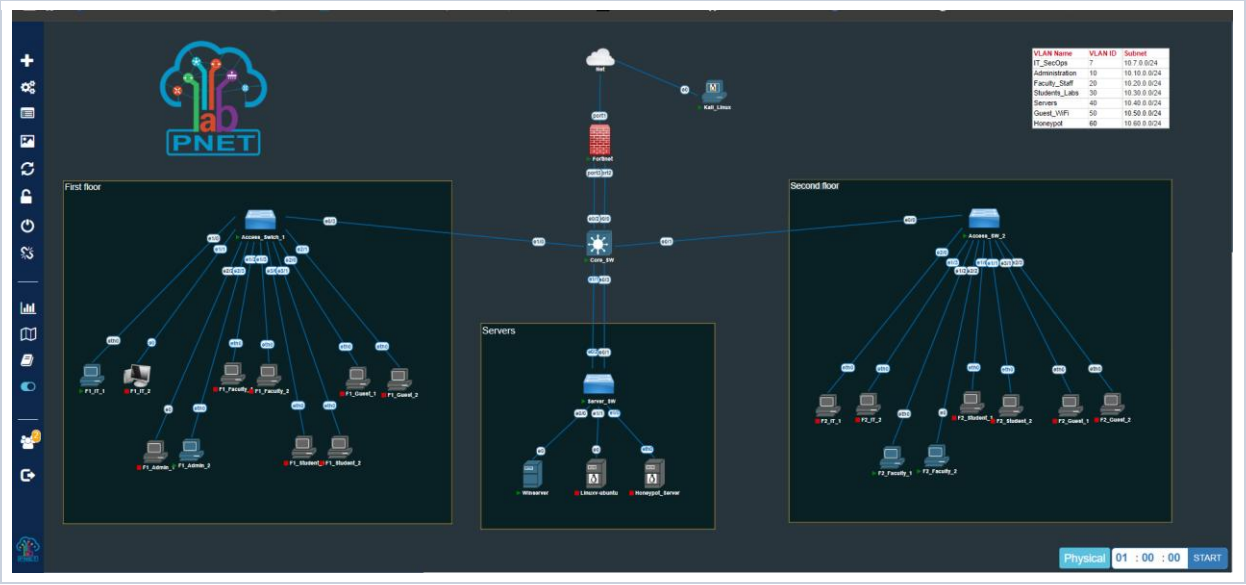


Figure 1.1: Overall network topology in the PNETLab environment — shows the main network devices, their distribution across the two floors and the server zone, with the VLAN table at the top of the figure.

The following table summarizes the physical links between network components:

Link	First End	Second End	Type
Core_SW ↔ FortiGate	Ethernet0/0 , Ethernet0/2	port2 , port3 (Core_LAG)	Port-channel 2 (Static)
Core_SW ↔ Server_SW	Ethernet1/1 , Ethernet0/3	Ethernet0/1 , Ethernet0/2	Port-channel 1 (LACP)
Core_SW ↔ Access_SW_1	Ethernet1/0	Ethernet0/3	Trunk 802.1Q

Link	First End	Second End	Type
Core_SW ↔ Access_SW_2	Ethernet0/1	Ethernet0/0	Trunk 802.1Q

1.6 VLAN Plan

The network relies on 7 isolated security zones, in addition to VLAN 444, which is reserved as a dummy Native VLAN, as shown in the following table:

VLAN ID	VLAN Name	Role	Subnet
7	IT_SecOps	IT and information security team, and the device management network.	10.7.0.0/24
10	Administration	Administration department.	10.10.0.0/24
20	Faculty_Staff	Faculty and teaching staff.	10.20.0.0/24
30	Students_Labs	Student labs.	10.30.0.0/24
40	Servers	Windows Server (AD) and Wazuh Manager.	10.40.0.0/24
50	Guest_WiFi	Guest network.	10.50.0.0/24
60	Honeypot	Isolated decoy (honeypot) server.	10.60.0.0/24
444	Native_Drop	Dummy Native VLAN with no active devices.	-

1.7 Report Structure

- Chapter 2: Foundation phase and Layer 2 build-out (naming, Spanning Tree, VLANs, EtherChannel, Trunks, VLAN pruning, VTP).
- Chapter 3: Layer 2 security (Port Security, BPDU Guard, DHCP Snooping, physical protection, server port protection).
- Chapter 4: Administrative device hardening (SSHv2, break-glass accounts, session security, disabling unnecessary services, login banner, Management VLAN interface).
- Chapter 5: Routing and integration with the FortiGate (Layer 3 and DHCP).
- Chapter 6: Strategic engineering decisions.
- Chapter 7: Simulation-environment constraints and real-world alternatives.
- Chapter 8: Conclusion and next steps.

Chapter 2: Foundation Phase and Infrastructure (Layer 2 Foundation)

This phase aims to build a stable, fast, and logically isolated network environment that guarantees an uninterrupted flow of data and logs. The following steps are applied to the four switches: Core_SW, Server_SW, Access_SW_1, and Access_SW_2. The `write memory` command is executed after each group of settings to save the configuration to the startup-config.

2.1 Device Naming (Hostnames)

Description	Assign a standard name to each switch that reflects its role in the network: Core_SW, Server_SW, Access_SW_1, and Access_SW_2.
Rationale	Consistent naming simplifies device management and pinpoints the exact source of any attack or security alert in the Wazuh logs.
Importance	When any log reaches Wazuh SIEM it can be tied immediately to the device concerned without guesswork, which raises the accuracy of security analysis and shortens incident response time.

Code

Cisco CLI | All Switches

```
! ===== Core_SW =====
configure terminal
hostname Core_SW
end
write memory
! ===== Server_SW =====
configure terminal
hostname Server_SW
end
write memory
! ===== Access_SW_1 =====
configure terminal
hostname Access_SW_1
end
write memory
! ===== Access_SW_2 =====
configure terminal
hostname Access_SW_2
end
write memory
```

The name of each device appears in the command prompt in all screenshots included in this report.

2.2 Rapid-PVST+ and Root Bridge Selection

Description	Enable the Rapid-PVST+ version of Spanning Tree on all switches and designate Core_SW as the Root Bridge for all VLANs using the `root primary` command. In the saved startup-config this command appears as `spanning-tree vlan 1-4094 priority 24576`, which the system converts automatically.
Rationale	Prevent endless loops that can cause broadcast storms and bring the network down. The rapid version restores the network in under two seconds after a topology change, while making Core_SW the root keeps traffic paths centered on the core of the network.
Importance	Preserves network stability and the uninterrupted flow of data and logs, and prevents an edge switch from being elected Root Bridge by mistake.

Code

Cisco CLI | Spanning Tree

```
! ===== Core_SW =====
configure terminal
spanning-tree mode rapid-pvst
spanning-tree vlan 1-4094 root primary
end
write memory
! ===== Server_SW / Access_SW_1 / Access_SW_2 =====
configure terminal
spanning-tree mode rapid-pvst
end
write memory
```

Verification: The output of `show spanning-tree summary` on Core_SW shows that Rapid-PVST mode is enabled and that Core_SW is the Root Bridge.

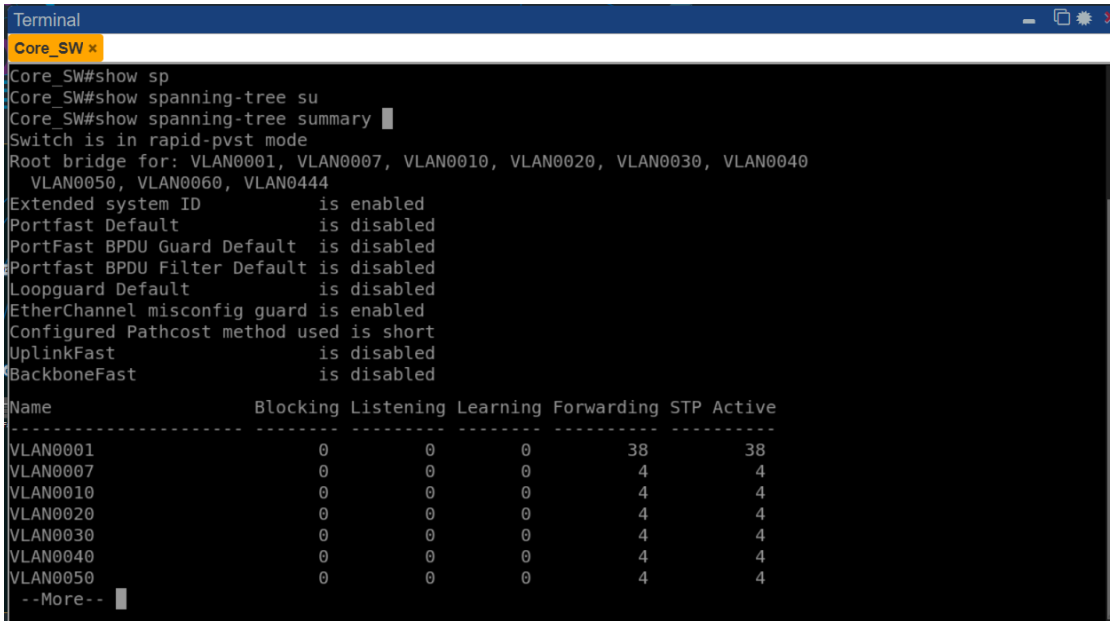


Figure 2.1: Output of `show spanning-tree summary` on Core_SW — proves that the device runs in rapid-pvst mode and is the Root Bridge for all VLANs.

2.3 VLAN Segmentation

Description	Create the VLANs on each switch according to the departments actually present on it only. Core_SW contains all VLANs; Server_SW contains VLAN 7 (for management, see Section 4.6), VLAN 40, and VLAN 60; Access_SW_1 contains VLAN 7, VLAN 10, VLAN 20, VLAN 30, and VLAN 50; Access_SW_2 does not contain VLAN 10 because the Administration department is not located on the second floor. VLAN 444 is added to all devices.
Rationale	Divide the network into 7 isolated security zones (such as Administration, Servers, and Guests) to reduce the attack surface, contain any compromise within its own zone, and prevent lateral movement. Creating on each switch only the VLANs it actually needs also applies the Zero-Trust principle.
Importance	Segmentation is the foundation on which firewall policies between departments are later built; without it, no inspection or restriction can be enforced on traffic between departments.

Code

Cisco CLI | Core_SW

```
configure terminal
! Create all security zones and the Native Drop VLAN
vlan 7
  name IT_SecOps
vlan 10
  name Administration
vlan 20
  name Faculty_Staff
vlan 30
  name Students_Labs
vlan 40
  name Servers
vlan 50
  name Guest_WiFi
vlan 60
  name Honeypot
vlan 444
  name Native_Drop
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
! VLAN 7 is used for switch management (see Section 4.6)
vlan 7
  name IT_SecOps
vlan 40
  name Servers
vlan 60
  name Honeypot
vlan 444
  name Native_Drop
end
write memory
```

Cisco CLI | Access_SW_1

```
configure terminal
vlan 7
```

Cisco CLI | Access_SW_1

```
  name IT_SecOps
vlan 10
  name Administration
vlan 20
  name Faculty_Staff
vlan 30
  name Students_Labs
vlan 50
  name Guest_WiFi
vlan 444
  name Native_Drop
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
! Only the VLANs present on the second floor
vlan 7
  name IT_SecOps
vlan 20
  name Faculty_Staff
vlan 30
  name Students_Labs
vlan 50
  name Guest_WiFi
vlan 444
  name Native_Drop
end
write memory
```

Verification: The `show vlan brief` command proves that the required VLANs were created on each switch, and confirms that Access_SW_2 does not contain unused networks such as the Administration network.

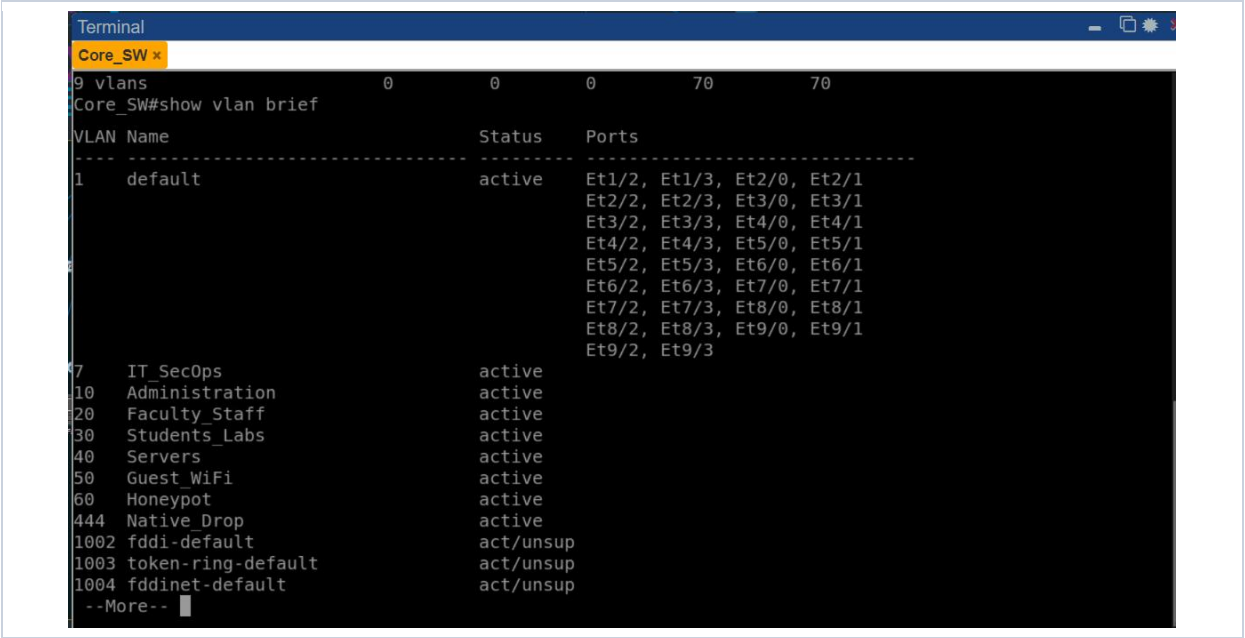


Figure 2.2: Output of `show vlan brief` on Core_SW — shows the creation of VLANs 7, 10, 20, 30, 40, 50, 60, and 444 on Core_SW.

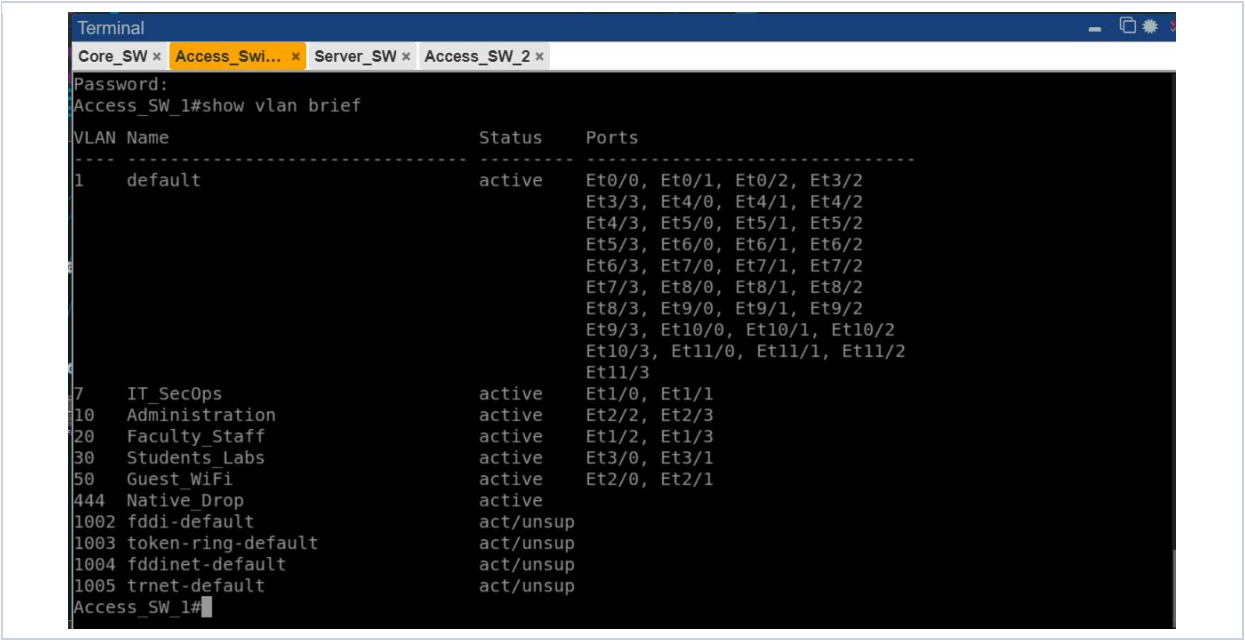


Figure 2.3: Output of `show vlan brief` on Access_SW_1 — shows the VLANs present on the first floor together with VLAN 444, and the placement of each department's ports in its own network.

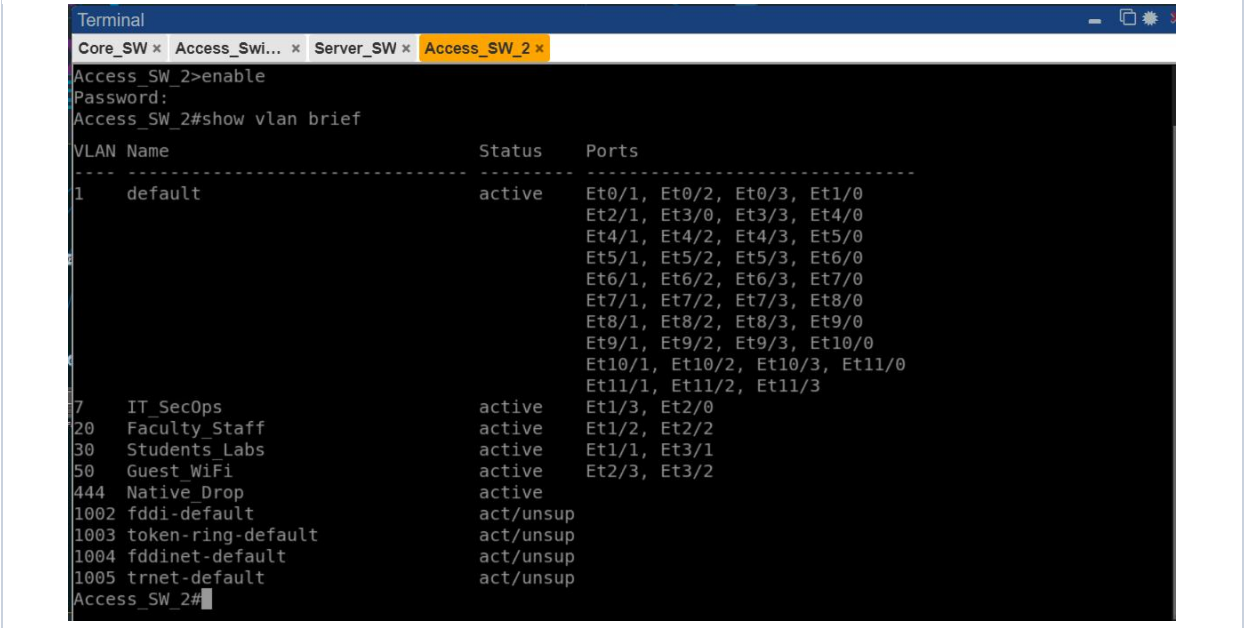


Figure 2.4: Output of `show vlan brief` on Access_SW_2 — shows the VLANs present on the second floor only; VLAN 10 (Administration) does not exist on this switch.

2.4 Changing the Native VLAN (Native VLAN 444)

Description	Change the default Native VLAN on all trunk links to VLAN 444, which contains no active devices, so that it acts as a “black hole” for any untagged traffic.
Rationale	Protect the main interconnect paths from VLAN hopping attacks, since an attacker finds no real network reachable through untagged frames.
Importance	Closes a well-known weakness in the default trunk configuration with no performance impact.

Code

Cisco CLI | Trunk interfaces (excerpt)

```
! Applied on every Trunk link (full configuration in Section 2.6)
interface Port-channel 2
  switchport trunk native vlan 444
```

This setting is verified with the `show interfaces trunk` command in Section 2.6, where the value 444 appears in the Native vlan column.

2.5 Strategic Link Aggregation (EtherChannel - Static & LACP)

Description	Bundle multiple physical links between core devices into a single logical channel (Port-channel): two links between Core_SW and the FortiGate (Port-channel 2) in static mode (Static / Mode ON), and two links between Core_SW and Server_SW (Port-channel 1) using the LACP negotiation protocol.
Rationale	Static aggregation was adopted with the FortiGate to work around a software limitation in the simulation environment that prevents LACP negotiation from succeeding (details in Chapter 7), while LACP is used between Core_SW and Server_SW.
Importance	Provides doubled capacity (2 Gbps) with load balancing and eliminates the single point of failure; if one cable fails, data and Wazuh logs continue to flow without any service interruption (High Availability).

Code

Cisco CLI | Core_SW

```
configure terminal
! Two links to FortiGate (Static - Mode ON)
interface range Ethernet0/0 , Ethernet0/2
  description Link_to_FortiGate
  no channel-protocol lacp
  channel-group 2 mode on
  exit
! Two links to Server_SW (LACP)
interface range Ethernet1/1 , Ethernet0/3
  description LACP_to_Server_SW
  channel-group 1 mode active
  exit
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
! LACP bundle from Core_SW, with Trunk and Native VLAN 444
interface range Ethernet0/1 - 2
  description LACP_from_Core
  channel-group 1 mode active
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  no shutdown
  exit
end
write memory
```

Verification: The output of `show etherchannel summary` shows that Port-channel 2 is in state SU (Layer 2, in use) with no protocol, and that Port-channel 1 is in state SU with the LACP protocol; member ports carry the flag P, meaning they are successfully bundled into the channel.

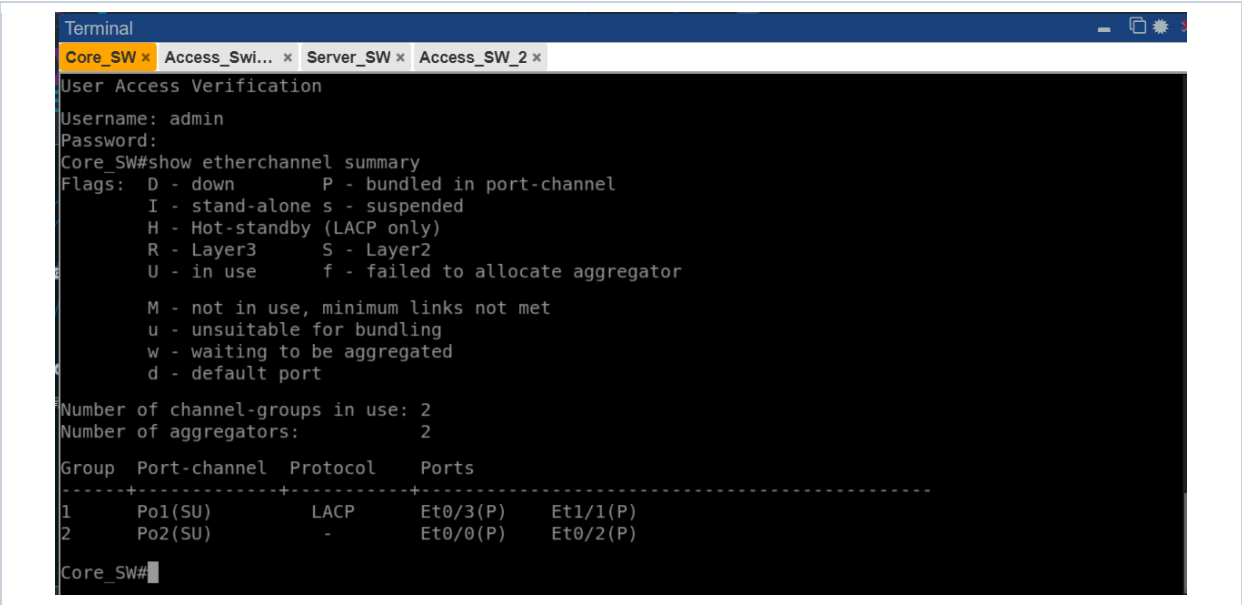


Figure 2.5: Output of `show etherchannel summary` on Core_SW — shows Port-channel 2 (static aggregation) and Port-channel 1 (LACP) in state SU, with member ports in state P.

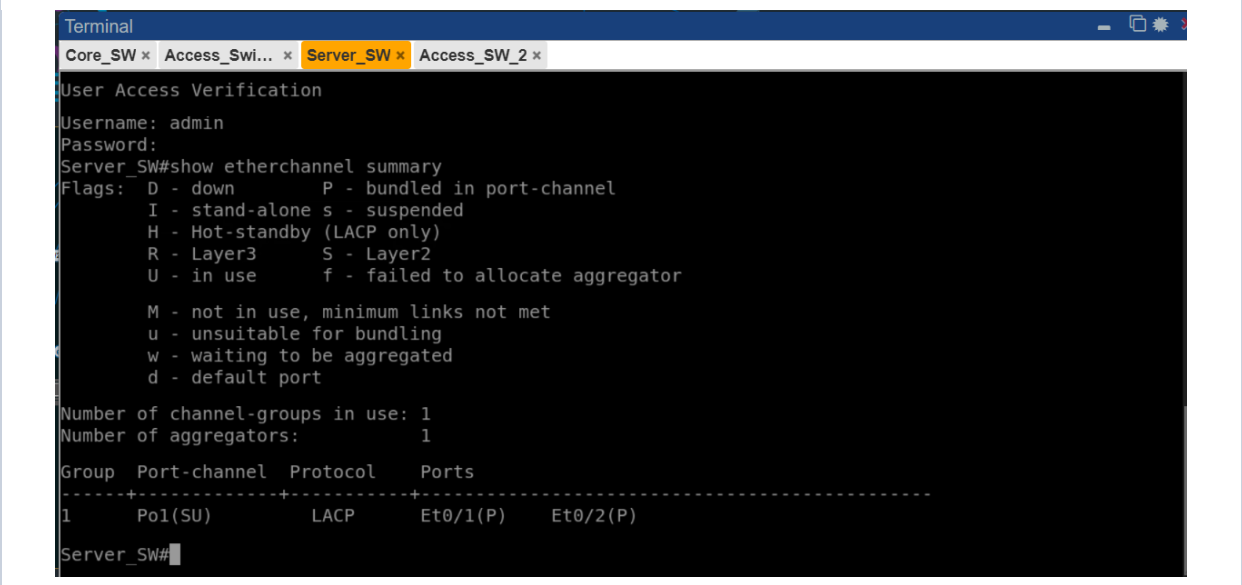


Figure 2.6: Output of `show etherchannel summary` on Server_SW — shows Port-channel 1 in state SU with the LACP protocol and its two bundled member ports Et0/1 and Et0/2.

2.6 Trunk Ports (Trunk 802.1Q)

Description	Configure inter-device links as trunks using 802.1Q to carry the traffic of all departments over a single organized path, while fixing Native VLAN 444 and disabling dynamic negotiation with the `switchport nonegotiate` command.
Rationale	Carry the traffic of different departments over one link instead of a dedicated link per department, while each department remains isolated inside its own VLAN.
Importance	Provides scalability and reduces cabling, while the accompanying security settings ensure that the link's function cannot change through dynamic negotiation. The Server_SW trunk was configured on the member ports in Section 2.5, then on Port-channel 1 itself as shown in the code.

Code

Cisco CLI | Core_SW

```
configure terminal
interface Port-channel 2
  description Trunk_to_FortiGate
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
interface Port-channel 1
  description Trunk_to_Server_SW
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
interface Ethernet1/0
  description Trunk_to_Access_SW_1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
interface Ethernet0/1
  description Trunk_to_Access_SW_2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
end
write memory
```

Cisco CLI | Access_SW_1

```
configure terminal
interface Ethernet0/3
  description Trunk_to_Core
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
```

Cisco CLI | Access_SW_1

```
  exit
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
interface Ethernet0/0
  description Trunk_to_Core
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
! Trunk configured on the Port-channel itself
interface Port-channel 1
  description Trunk_to_Core
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 444
  switchport nonegotiate
  no shutdown
  exit
end
write memory
```

Verification: The `show interfaces trunk` command confirms that all interconnect ports operate in Trunk mode using 802.1Q, and that the Native vlan column shows 444.

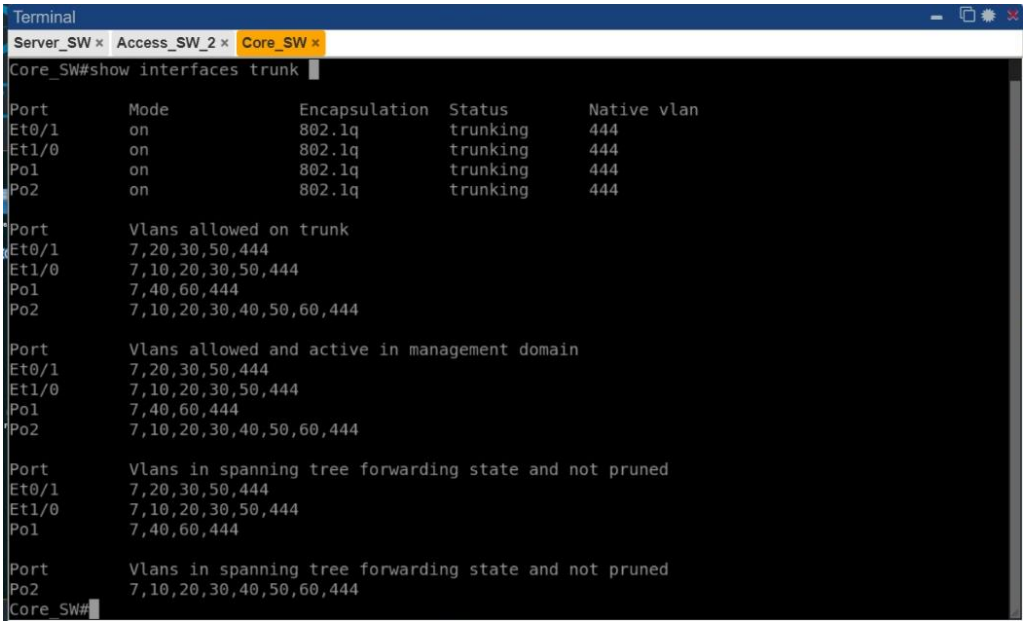


Figure 2.7: Output of `show interfaces trunk` on Core_SW — shows that Core_SW links operate as 802.1Q trunks with Native VLAN 444 and a restricted allowed-VLAN list on each link.

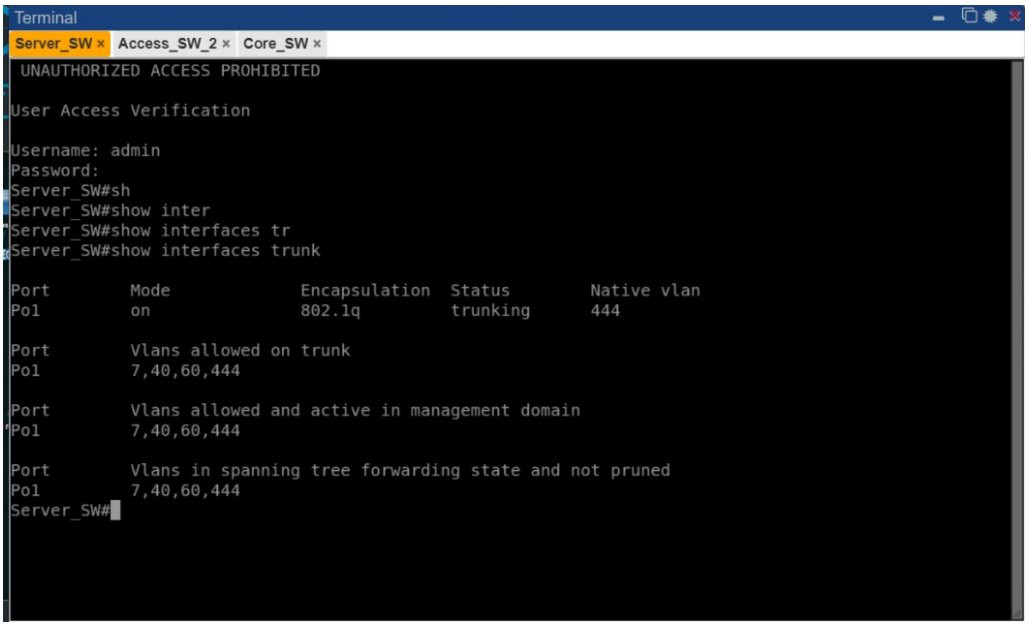


Figure 2.8: Output of `show interfaces trunk` on Server_SW — shows that Port-channel 1 operates as a trunk and carries only VLANs 7, 40, 60, and 444.

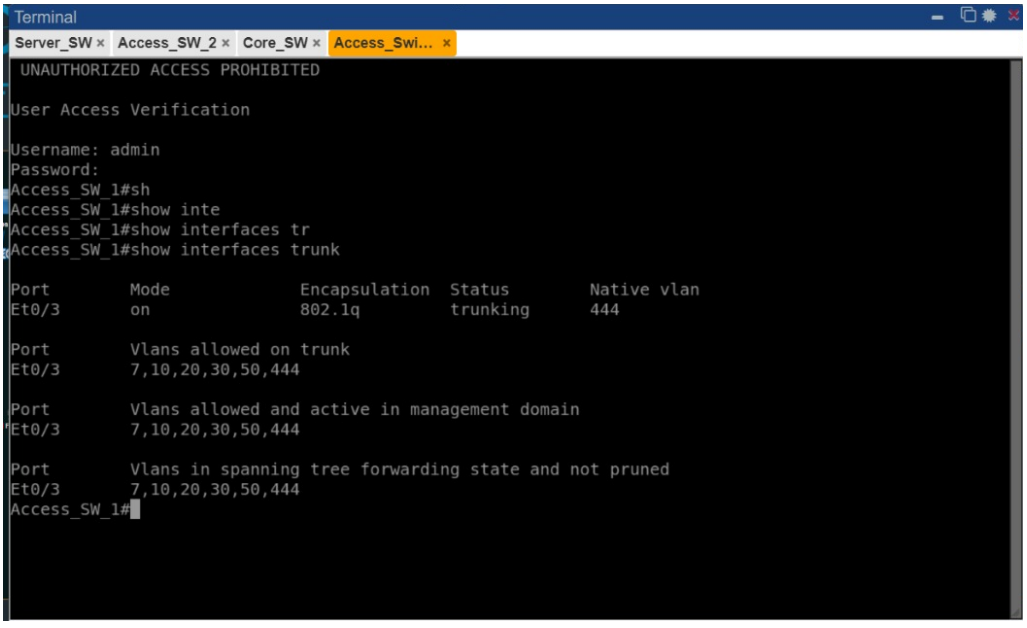


Figure 2.9: Output of `show interfaces trunk` on Access_SW_1 — shows that link Et0/3 carries only VLANs 7, 10, 20, 30, 50, and 444.

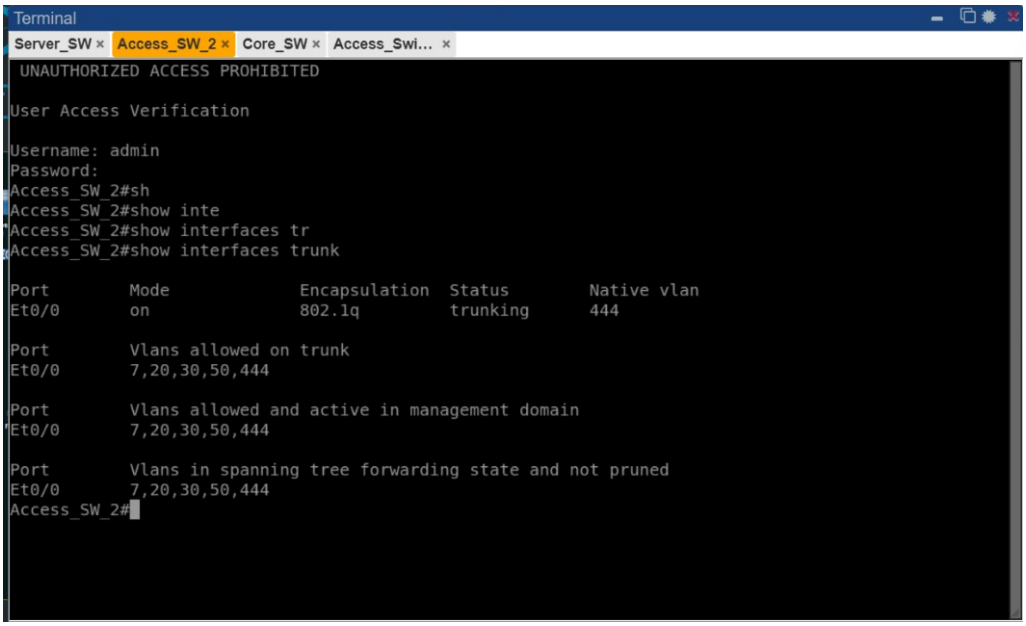


Figure 2.10: Output of `show interfaces trunk` on Access_SW_2 — shows that link Et0/0 carries only VLANs 7, 20, 30, 50, and 444.

2.7 Accelerating Edge Ports (PortFast)

Description	Enable `spanning-tree portfast` on edge-device ports. On Server_SW this includes the Windows Server (AD) port and the Wazuh Manager port in VLAN 40, and the Honeypot port in VLAN 60. User ports on Access_SW_1 and Access_SW_2 are configured in Chapter 3.
Rationale	Skip the Listening / Learning stages of Spanning Tree so the port comes up as soon as the cable is connected, preventing initial connectivity loss and problems obtaining an IP address.
Importance	Ensures that servers and edge devices are ready to communicate as soon as they power up, and that DHCP works properly.

Code

Cisco CLI | Server_SW

```
configure terminal
! Windows Server (Active Directory) - VLAN 40
interface Ethernet0/0
  description Windows_Server_AD
  switchport mode access
  switchport access vlan 40
  spanning-tree portfast
  no shutdown
  exit
! Wazuh SIEM Manager - VLAN 40
interface Ethernet1/1
  description Wazuh_Manager
  switchport mode access
  switchport access vlan 40
  spanning-tree portfast
  no shutdown
  exit
! Isolated Honeypot decoy - VLAN 60
interface Ethernet1/0
  description Honeypot_Decoy
  switchport mode access
```

Cisco CLI | Server_SW

```
  switchport access vlan 60
  spanning-tree portfast
  no shutdown
  exit
end
write memory
```

Verification: The `show vlan brief` command lists ports e0/0 and e1/1 in VLAN 40 and port e1/0 in VLAN 60, and `show spanning-tree summary` confirms that Server_SW runs in Rapid-PVST mode and that VLAN 40, VLAN 60, and VLAN 444 are in the Forwarding state.

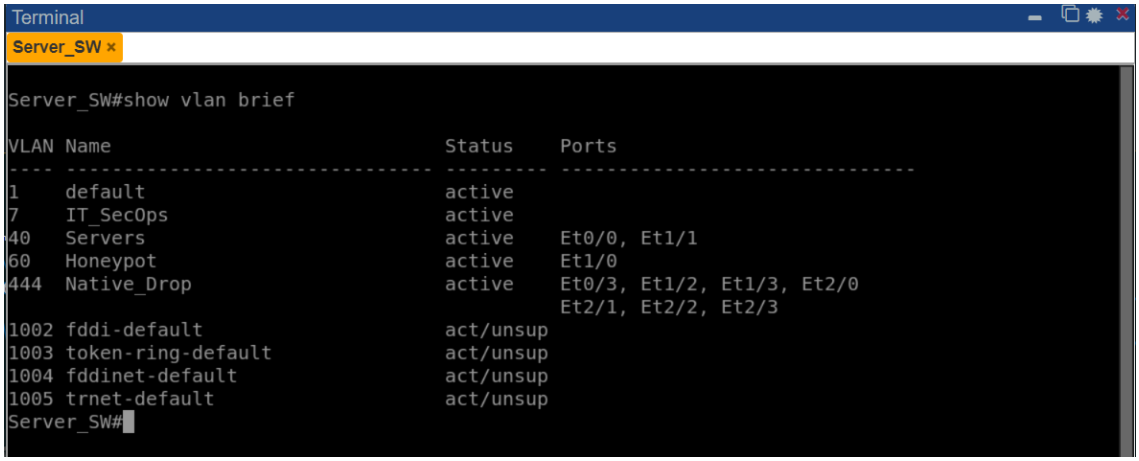


Figure 2.11: Output of `show vlan brief` on Server_SW — shows the creation of VLAN 7 for management, the placement of server ports (Et0/0 and Et1/1 in VLAN 40, Et1/0 in VLAN 60), and the unused ports parked in VLAN 444.

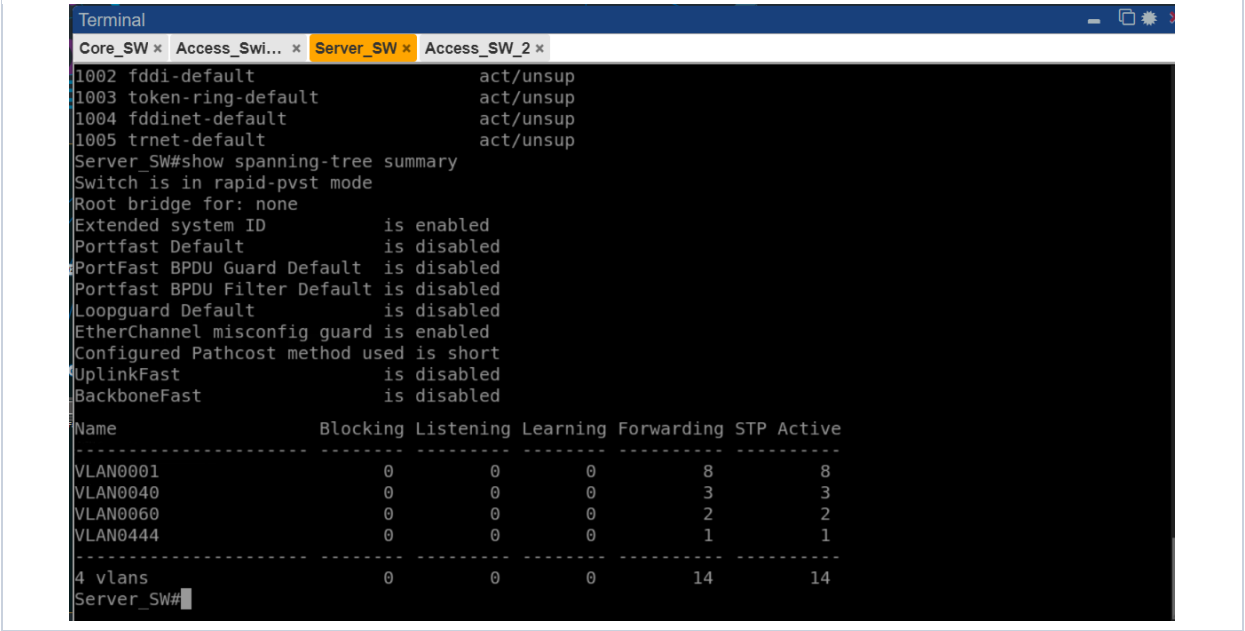


Figure 2.12: Output of `show spanning-tree summary` on Server_SW — shows that Server_SW runs in rapid-pvst mode, is not the Root Bridge, and has VLANs 40, 60, and 444 in the Forwarding state.

2.8 Restricting Allowed VLANs on Trunks (Trunk Pruning - Allowed VLAN)

Description	Specify the VLANs permitted to cross each trunk link using the `switchport trunk allowed vlan` command, so that each link carries only the networks it needs, while keeping VLAN 444 as the Native VLAN. The following table shows the list applied to each link.
Rationale	The default trunk configuration allows all VLANs from 1-4094, meaning broadcast traffic reaches networks that do not exist behind the link and any network created later by mistake automatically gains a path. Restricting the list applies the Least Privilege principle at Layer 2 and makes each department's path specific and documented.
Importance	Confines broadcast domains to the relevant devices only: the Administration network (VLAN 10) does not reach Access_SW_2, user networks do not reach Server_SW, and VLAN 1 does not cross the main links. This reduces the attack surface and keeps the design consistent with the Zero-Trust model.

Interface	Direction	Allowed VLANs
Port-channel 2	Core_SW → FortiGate	7,10,20,30,40,50,60,444
Port-channel 1	Core_SW → Server_SW	7,40,60,444
Ethernet1/0	Core_SW → Access_SW_1	7,10,20,30,50,444
Ethernet0/1	Core_SW → Access_SW_2	7,20,30,50,444
Port-channel 1	Server_SW → Core_SW	7,40,60,444
Ethernet0/3	Access_SW_1 → Core_SW	7,10,20,30,50,444
Ethernet0/0	Access_SW_2 → Core_SW	7,20,30,50,444

Code

Cisco CLI | Core_SW

```
configure terminal
interface Port-channel 2
  switchport trunk allowed vlan 7,10,20,30,40,50,60,444
  exit
interface Port-channel 1
  switchport trunk allowed vlan 7,40,60,444
  exit
interface Ethernet1/0
  switchport trunk allowed vlan 7,10,20,30,50,444
  exit
interface Ethernet0/1
  switchport trunk allowed vlan 7,20,30,50,444
  exit
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
interface Port-channel 1
```

Cisco CLI | Server_SW

```
  switchport trunk allowed vlan 7,40,60,444
  exit
end
write memory
```

Cisco CLI | Access_SW_1

```
configure terminal
interface Ethernet0/3
  switchport trunk allowed vlan 7,10,20,30,50,444
  exit
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
interface Ethernet0/0
  switchport trunk allowed vlan 7,20,30,50,444
  exit
end
write memory
```

Verification: The `show interfaces trunk` command documented in Figures 2.7 to 2.10 confirms that the Vlans allowed on trunk column shows the restricted lists in the table above instead of 1-4094.

2.9 Disabling VLAN Propagation (VTP Transparent Mode)

Description	Set VTP to Transparent mode on all switches using the `vtp mode transparent` command, so that each switch relies only on its own local VLAN database, while VTP messages continue to be forwarded to other devices.
Rationale	VTP propagates the VLAN database automatically between devices. In the default Server mode, connecting a switch with a newer database, or deleting a VLAN by mistake, can overwrite the database on every device and take the network down. Transparent mode prevents this scenario.
Importance	Protects the VLAN design from accidental or malicious modification, and is consistent with the principle that each switch only creates the networks actually present on it, as described in Section 2.3.

Code

Cisco CLI | All Switches

```
configure terminal
! Core_SW, Server_SW, Access_SW_1, Access_SW_2
vtp mode transparent
end
write memory
```

Verification: The `show vtp status` command shows that the VTP Operating Mode field holds the value Transparent.

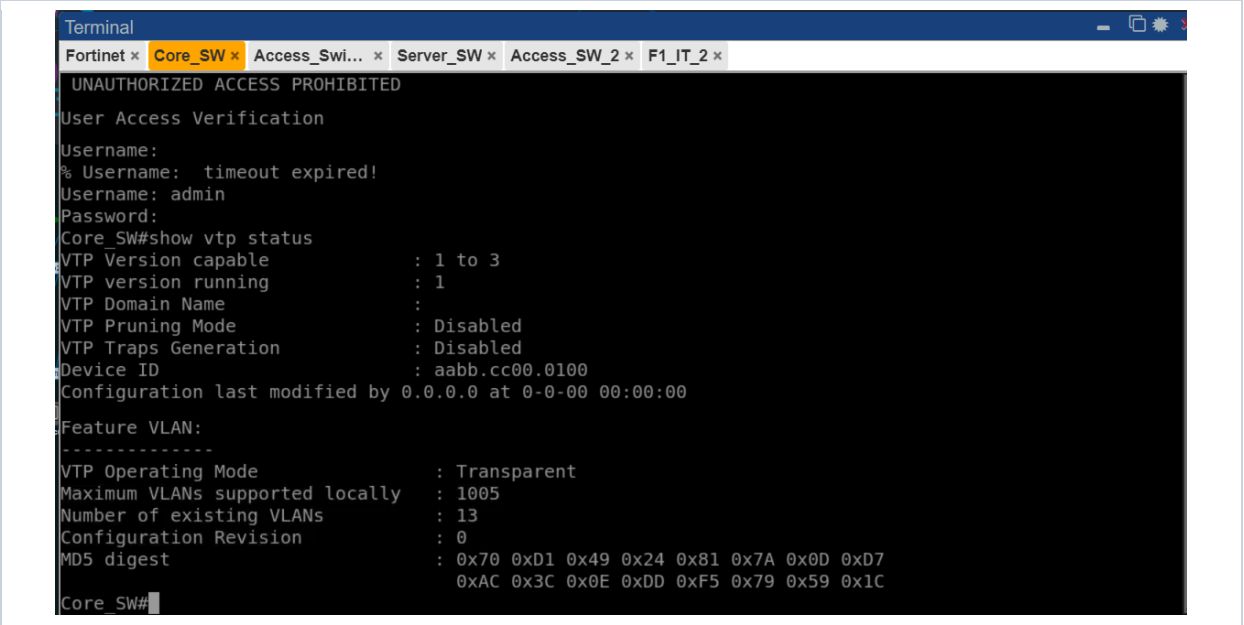


Figure 2.13: Output of `show vtp status` on Core_SW — shows that the VTP Operating Mode field holds the value Transparent.

Chapter 3: Layer 2 Security and Hardening

This phase secures the edge user ports (Access Ports) as the first line of defense against internal threats, since any compromised device inside the network starts its attack from one of these ports. Protection is applied to Access_SW_1 and Access_SW_2, and completed on Core_SW and Server_SW in the DHCP Snooping step.

3.1 Hardening User Access Ports

Description	A unified set of settings is applied to every user port: the port is forced into Access mode and placed in its department's VLAN, then surrounded by four layers of protection shown in the following table.
Rationale	The edge port is the first point of contact between the network and the user, and therefore the most likely entry point for any internal attack; its behavior must be fully restricted.
Importance	This set forms a complete security wall that prevents the port from becoming a trunk, limits it to a single device, protects the Root Bridge, and avoids any delay in bringing the port up.

Technology and Command	Rationale	Importance
DTP - Nonegotiate `switchport nonegotiate`	Frustrate attempts by attackers to convert an ordinary user port into a trunk to reach the other isolated networks.	Prevents the port's function from changing through dynamic negotiation (DTP).

Technology and Command	Rationale	Importance
Port Security `maximum 1` `mac-address sticky` `violation shutdown`	Limit the connection to a single device per port while recording its MAC address. The switch automatically shuts the port down (Violation Shutdown) as soon as an unauthorized device or an external switch is detected.	Prevents the connection of unknown devices and fake MAC address generation tools, and shuts the port down immediately upon violation.
BPDU Guard `spanning-tree bpduguard enable`	Prevent any compromised device in student labs or staff offices from sending BPDU messages in an attempt to hijack the Root Bridge role and destroy the network topology.	Protects the Root Bridge position on Core_SW and maintains Spanning Tree stability.
PortFast `spanning-tree portfast`	Skips the Listening and Learning stages to bring the port up as soon as the cable is connected.	Prevents initial connectivity loss and problems obtaining IP addresses.

The following table shows the distribution of ports across departments on each switch:

Device	VLAN	Department	Ports
Access_SW_1	7	IT_SecOps	Ethernet1/0 - 1
Access_SW_1	10	Administration	Ethernet2/2 - 3
Access_SW_1	20	Faculty_Staff	Ethernet1/2 - 3
Access_SW_1	30	Students_Labs	Ethernet3/0 - 1
Access_SW_1	50	Guest_WiFi	Ethernet2/0 - 1
Access_SW_2	7	IT_SecOps	Ethernet2/0 , Ethernet1/3
Access_SW_2	20	Faculty_Staff	Ethernet1/2 , Ethernet2/2
Access_SW_2	30	Students_Labs	Ethernet1/0 - 1 , Ethernet3/1
Access_SW_2	50	Guest_WiFi	Ethernet2/3 , Ethernet3/2

Code

Cisco CLI | Access_SW_1

```
configure terminal
! IT_SecOps ports - VLAN 7
interface range Ethernet1/0 - 1
  description IT_Access
  switchport mode access
  switchport access vlan 7
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
```

Cisco CLI | Access_SW_1

```
  switchport port-security violation shutdown
  no shutdown
  exit
! Administration ports - VLAN 10
interface range Ethernet2/2 - 3
  description Admin_Access
  switchport mode access
  switchport access vlan 10
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Faculty_Staff ports - VLAN 20
interface range Ethernet1/2 - 3
  description Faculty_Access
  switchport mode access
  switchport access vlan 20
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Students_Labs ports - VLAN 30
interface range Ethernet3/0 - 1
  description Students_Access
  switchport mode access
  switchport access vlan 30
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Guest_WiFi ports - VLAN 50
interface range Ethernet2/0 - 1
  description Guest_Access
  switchport mode access
  switchport access vlan 50
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
! IT_SecOps ports - VLAN 7
interface range Ethernet2/0 , Ethernet1/3
  description IT_Access
```

Cisco CLI | Access_SW_2

```
  switchport mode access
  switchport access vlan 7
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Faculty_Staff ports - VLAN 20
interface range Ethernet1/2 , Ethernet2/2
  description Faculty_Access
  switchport mode access
  switchport access vlan 20
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Students_Labs ports - VLAN 30
interface range Ethernet1/0 - 1 , Ethernet3/1
  description Students_Access
  switchport mode access
  switchport access vlan 30
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
! Guest_WiFi ports - VLAN 50
interface range Ethernet2/3 , Ethernet3/2
  description Guest_Access
  switchport mode access
  switchport access vlan 50
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  no shutdown
  exit
end
write memory
```

Verification: The `show port-security` command displays the port security table showing the maximum of one device per port and the Shutdown action on violation, and `show vlan brief` proves that each port is assigned to its own department.

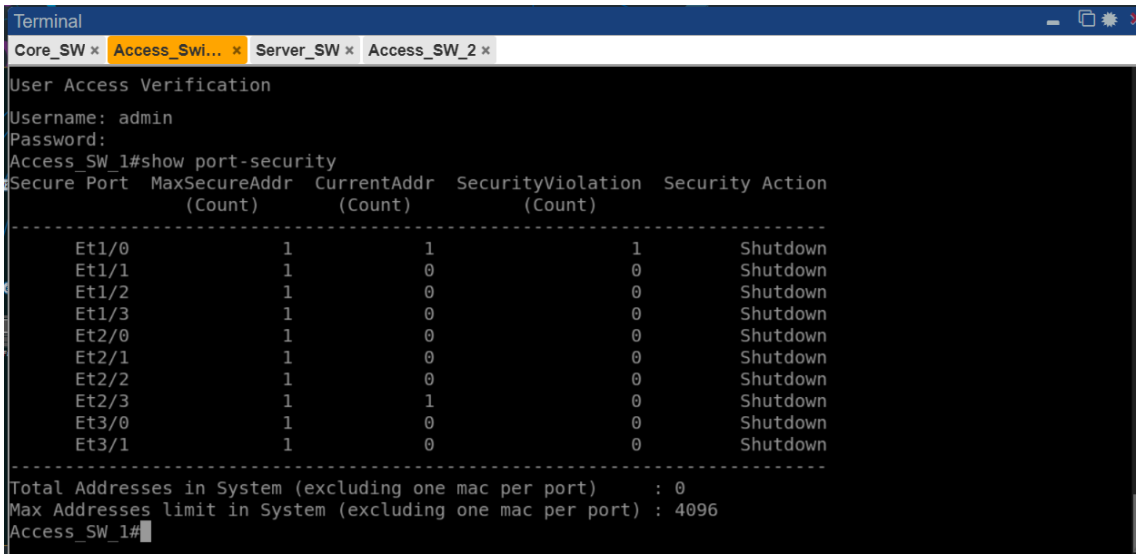


Figure 3.1: Output of `show port-security` on Access_SW_1 — shows Port Security enabled on user ports with a maximum of one device per port and a Shutdown action on violation.

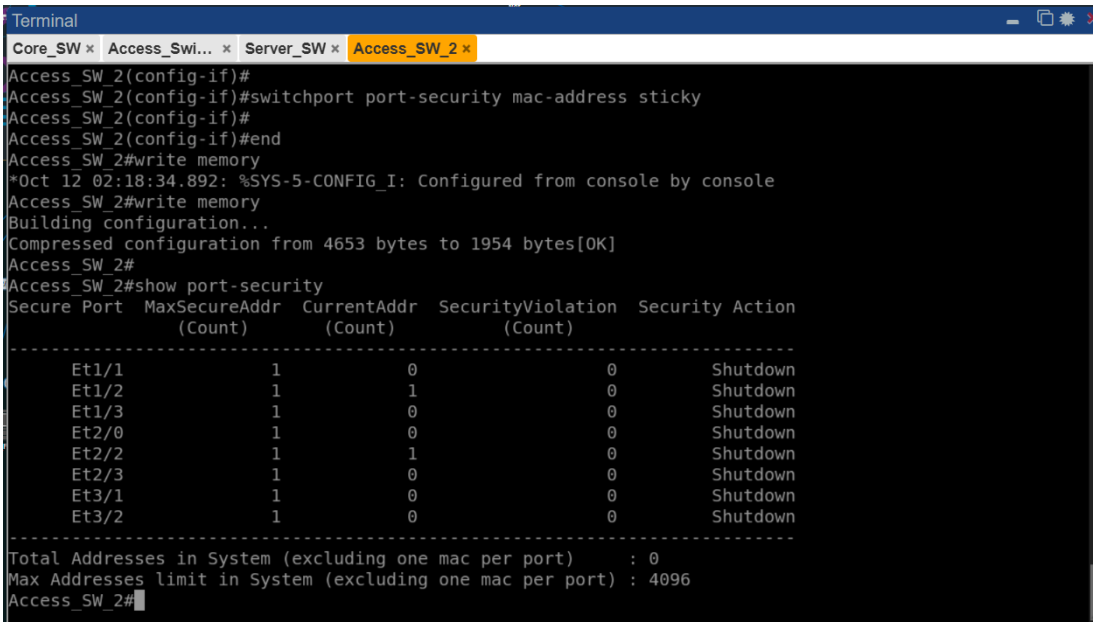


Figure 3.2: Output of `show port-security` on Access_SW_2 — shows the secured ports on Access_SW_2 with a maximum of one device per port and a Shutdown action on violation.

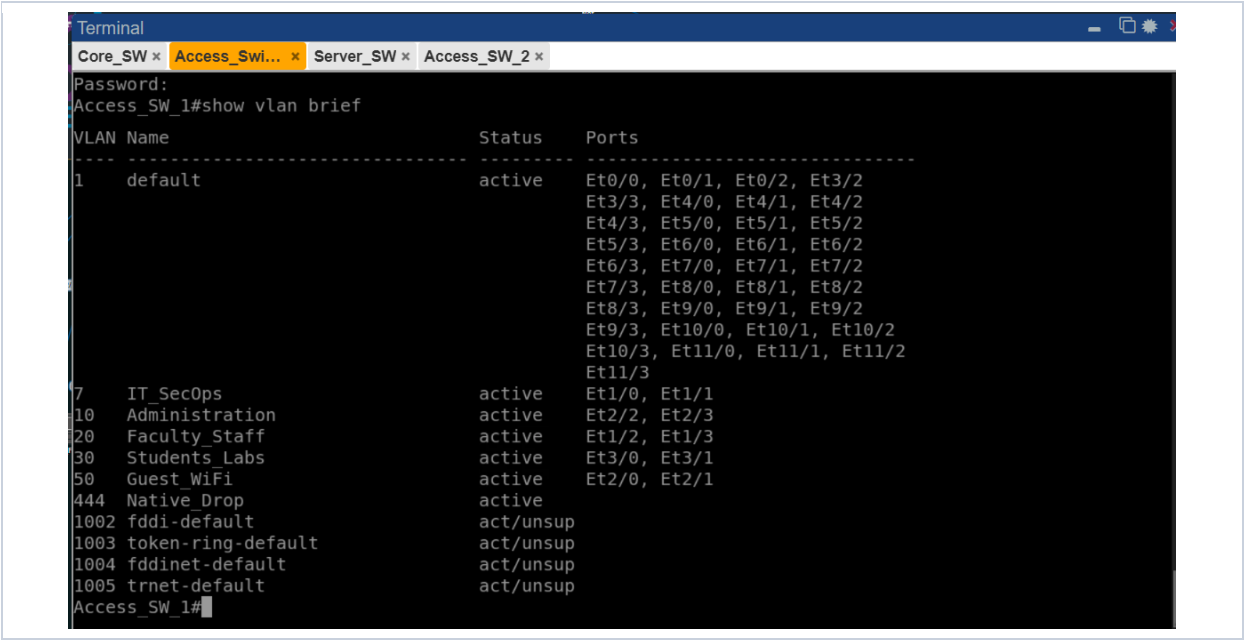


Figure 3.3: Output of `show vlan brief` on Access_SW_1 — shows each Access_SW_1 port placed in the VLAN of its own department.

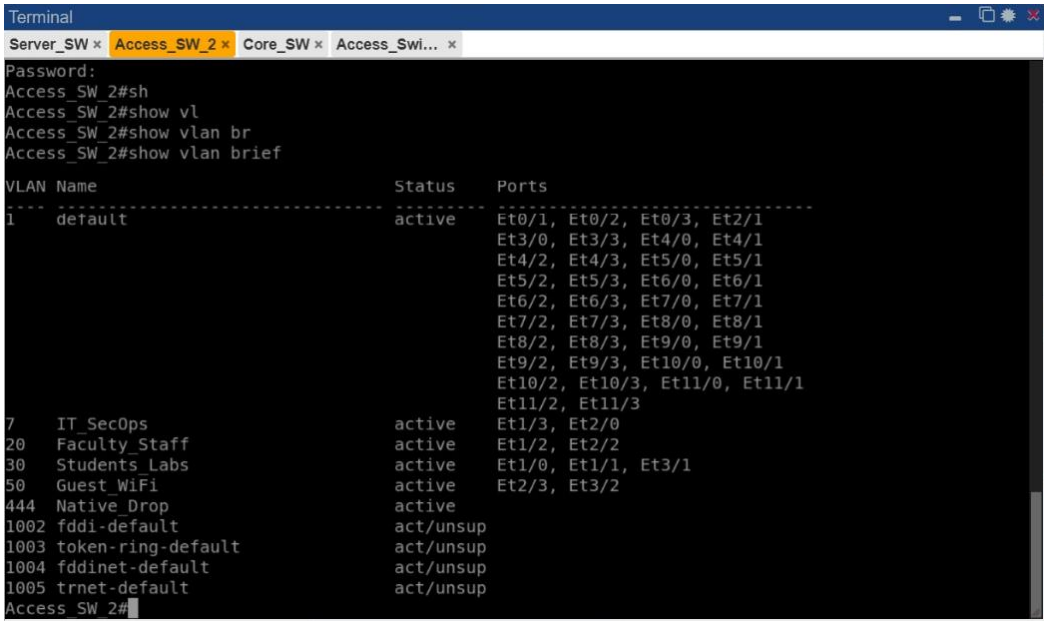


Figure 3.4: Output of `show vlan brief` on Access_SW_2 — shows the IT, Faculty, Students, and Guest ports on Access_SW_2, including the additional Students port Et1/0.

3.2 Address Security and Trust Boundaries (DHCP Snooping & Trusted Ports)

Description	Enable DHCP Snooping on all switches for the VLANs present on them, and designate exactly one trusted port on each device: the port or bundled path facing the FortiGate: Ethernet0/3 on Access_SW_1, Ethernet0/0 on Access_SW_2, Port-channel 2 on Core_SW, and Port-channel 1 on Server_SW. All remaining ports stay untrusted by default.
Rationale	Prevent rogue DHCP server attacks that distribute malicious default gateways to steal data, by confining trust exclusively to the path coming from the FortiGate so that it is the only legitimate distributor of IP addresses.
Importance	Protects edge devices from having their traffic routed through a fake gateway, strengthens central control over address distribution, and prevents address spoofing attacks. It also completes the protection scheme that forces the network to accept IP addresses from the FortiGate only.

Code

Cisco CLI | Access_SW_1

```
configure terminal
ip dhcp snooping
ip dhcp snooping vlan 7,10,20,30,50
no ip dhcp snooping information option
! Trust only the path that leads to the FortiGate
interface Ethernet0/3
  ip dhcp snooping trust
  exit
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
! Snooping is applied only to the VLANs present on this Switch
ip dhcp snooping
ip dhcp snooping vlan 7,20,30,50
no ip dhcp snooping information option
! Trust only the path that leads to the FortiGate
interface Ethernet0/0
  ip dhcp snooping trust
  exit
end
write memory
```

Cisco CLI | Core_SW

```
configure terminal
ip dhcp snooping
ip dhcp snooping vlan 7,10,20,30,40,50,60
no ip dhcp snooping information option
! Trust only the path that leads to the FortiGate
interface Port-channel 2
  ip dhcp snooping trust
  exit
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
ip dhcp snooping
ip dhcp snooping vlan 40,60
no ip dhcp snooping information option
! Trust only the path that leads to the FortiGate
interface Port-channel 1
  ip dhcp snooping trust
  exit
end
write memory
```

Verification: The `show ip dhcp snooping` command shows that inspection is enabled on the specified VLANs and that the listed port is the only Trusted port on each switch.

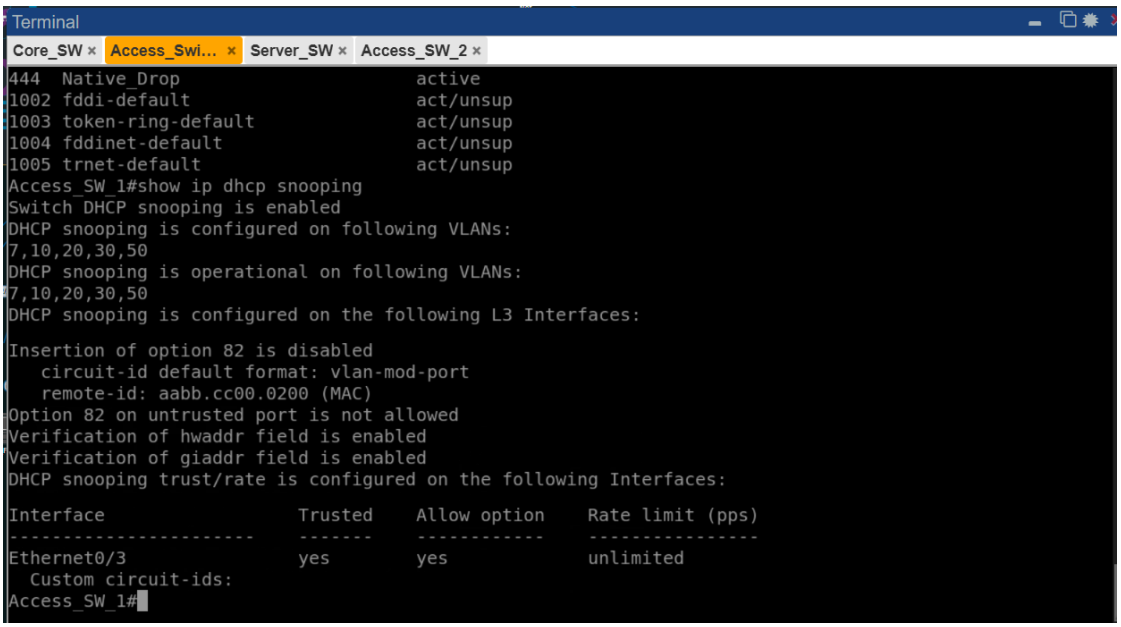


Figure 3.5: Output of `show ip dhcp snooping` on Access_SW_1 — shows DHCP Snooping enabled on VLANs 7, 10, 20, 30, and 50, with Ethernet0/3 as the only trusted port.

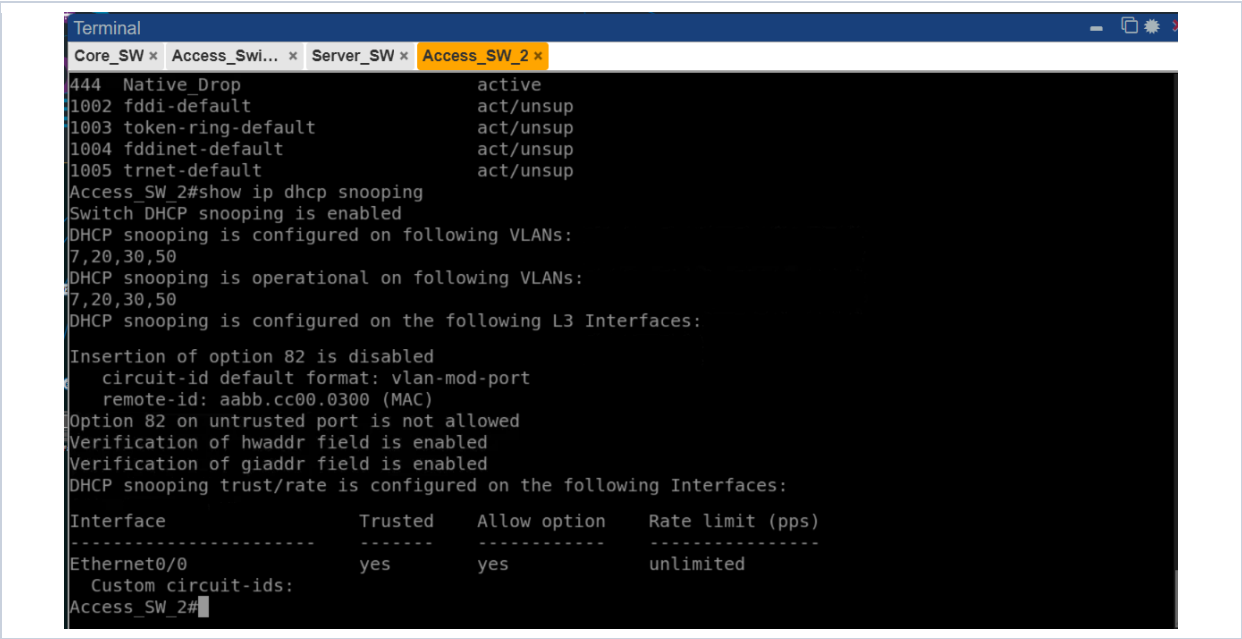


Figure 3.6: Output of `show ip dhcp snooping` on Access_SW_2 — shows DHCP Snooping enabled on VLANs 7, 20, 30, and 50, with Ethernet0/0 as the only trusted port.

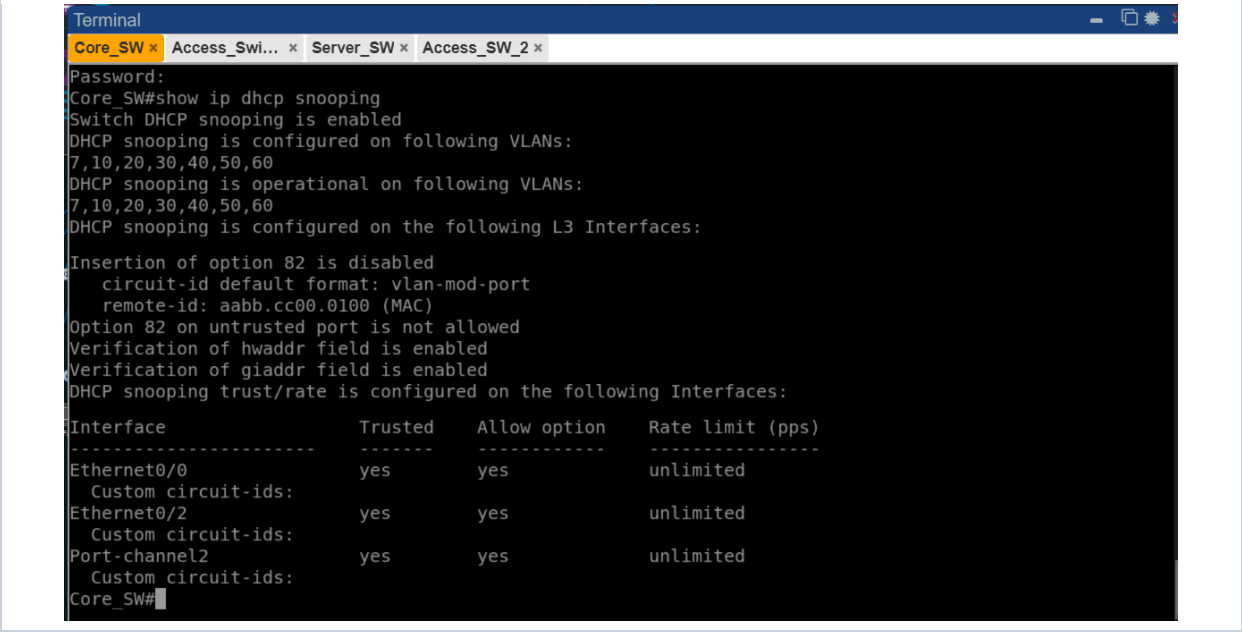


Figure 3.7: Output of `show ip dhcp snooping` on Core_SW — shows DHCP Snooping enabled on Core_SW, with Port-channel 2 toward the FortiGate as the trusted path.

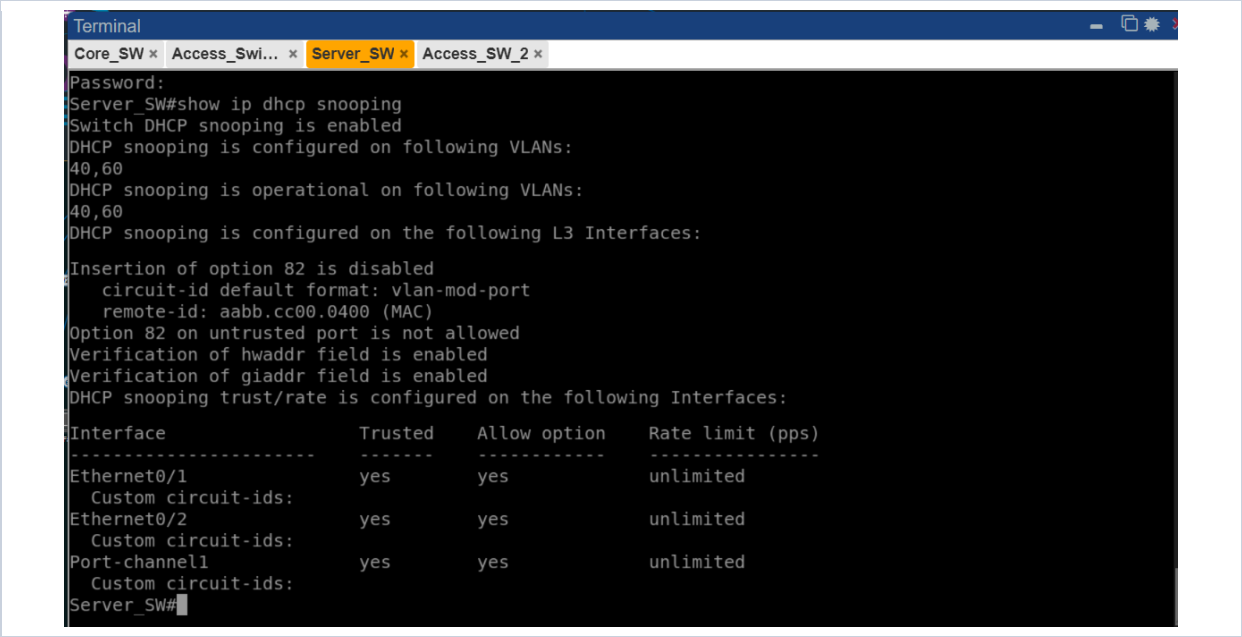


Figure 3.8: Output of `show ip dhcp snooping` on Server_SW — shows DHCP Snooping enabled on VLANs 40 and 60, with Port-channel 1 as the trusted path.

3.3 Physical Intrusion Prevention

Description	Permanently shut down (`shutdown`) any unused network port, place it in VLAN 444 (the dummy network with no devices), and name it Unused_Ports. This covers: Ethernet0/0 - 2, Ethernet3/2 - 3, and Ethernet4/0 - 3 on Access_SW_1; Ethernet0/1 - 3, Ethernet2/1, Ethernet3/0, Ethernet3/3, and Ethernet4/0 - 3 on Access_SW_2; Ethernet1/2 - 3, Ethernet2/0 - 3, and Ethernet3/0 - 3 on Core_SW; and Ethernet0/3, Ethernet1/2 - 3, and Ethernet2/0 - 3 on Server_SW.
Rationale	Prevent intruders from physically connecting external devices to the network through empty wall or lab ports. Placing the port in VLAN 444 also ensures that even if the port is re-enabled by mistake, any device connecting to it lands in an isolated network with no gateway and no services.
Importance	Closes the entry point for physical attacks and reduces the number of potential entry points to the minimum, through two consecutive layers of protection: administrative shutdown and placement in the dummy network.

Code

Cisco CLI | Access_SW_1

```
configure terminal
! Shut down all unused ports and park them in the Native Drop VLAN
interface range Ethernet0/0 - 2 , Ethernet3/2 - 3 , Ethernet4/0 - 3
  description Unused_Ports
  switchport mode access
  switchport access vlan 444
  shutdown
  exit
end
write memory
```

Cisco CLI | Access_SW_2

```
configure terminal
! Shut down all unused ports and park them in the Native Drop VLAN
interface range Ethernet0/1 - 3 , Ethernet2/1 , Ethernet3/0 , Ethernet3/3 , Ethernet4/0
- 3
  description Unused_Ports
  switchport mode access
  switchport access vlan 444
  shutdown
  exit
end
write memory
```

Cisco CLI | Core_SW

```
configure terminal
! Shut down all unused ports and park them in the Native Drop VLAN
interface range Ethernet1/2 - 3 , Ethernet2/0 - 3 , Ethernet3/0 - 3
  description Unused_Ports
  switchport mode access
  switchport access vlan 444
  shutdown
  exit
end
write memory
```

Cisco CLI | Server_SW

```
configure terminal
! Shut down all unused ports and park them in the Native Drop VLAN
interface range Ethernet0/3 , Ethernet1/2 - 3 , Ethernet2/0 - 3
  description Unused_Ports
  switchport mode access
  switchport access vlan 444
  shutdown
  exit
end
write memory
```

Verification: The `show ip interface brief` command shows that all the ports listed above are administratively down.

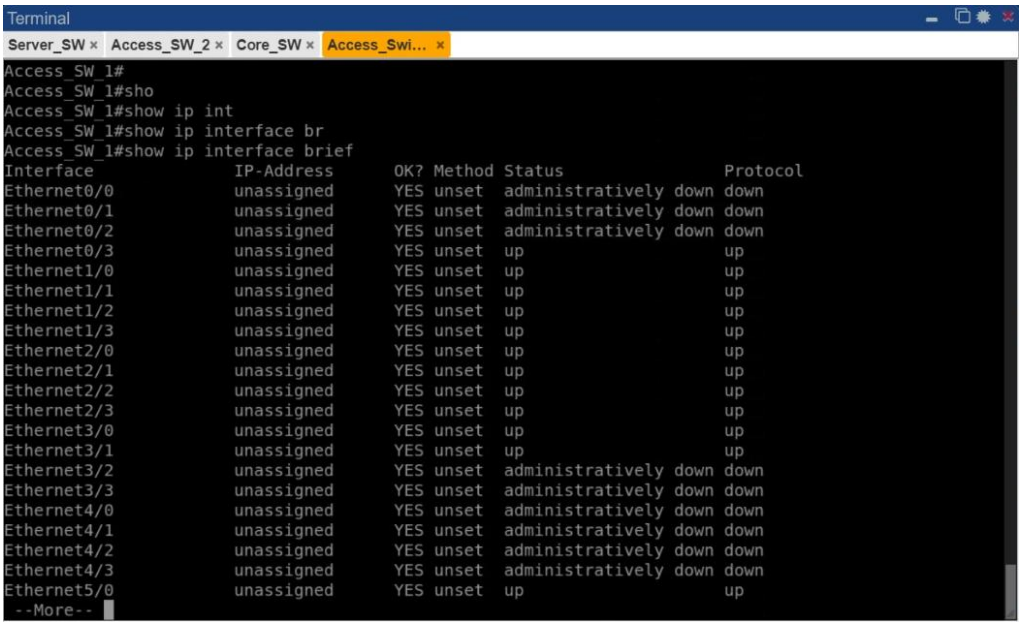


Figure 3.9: Output of `show ip interface brief` on Access_SW_1 — shows that the unused ports Et0/0-2, Et3/2-3, and Et4/0-3 are administratively down.

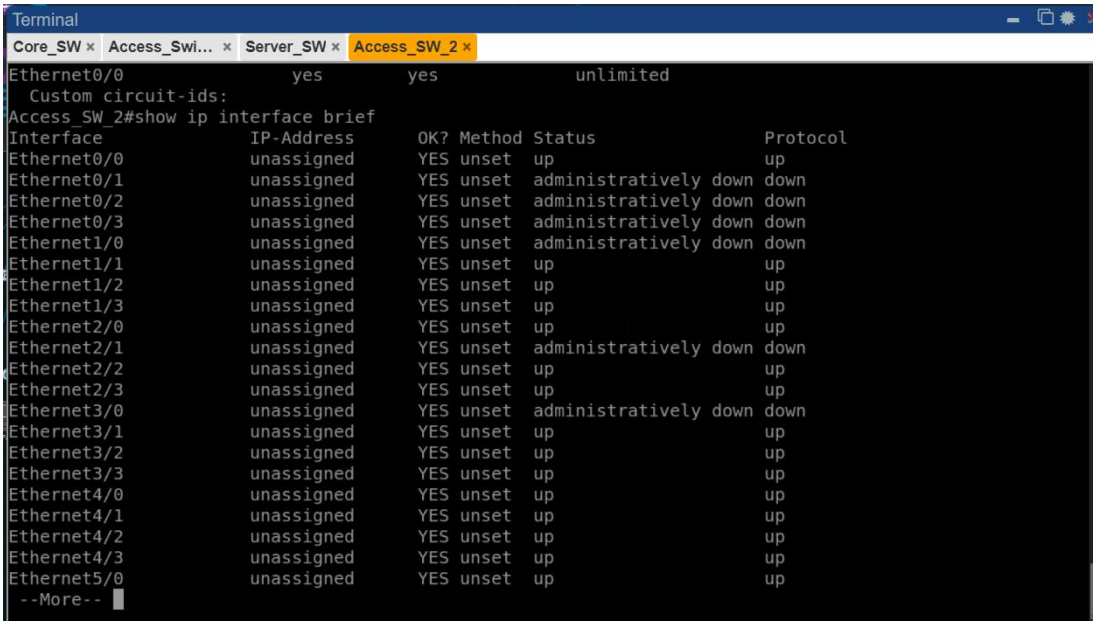


Figure 3.10: Output of `show ip interface brief` on Access_SW_2 — shows the port status on Access_SW_2 and that the unused ports are administratively down.



Figure 3.11: Output of `show ip interface brief` on Core_SW — shows the unused ports on Core_SW administratively down, and the Vlan7 interface with address 10.7.0.2.

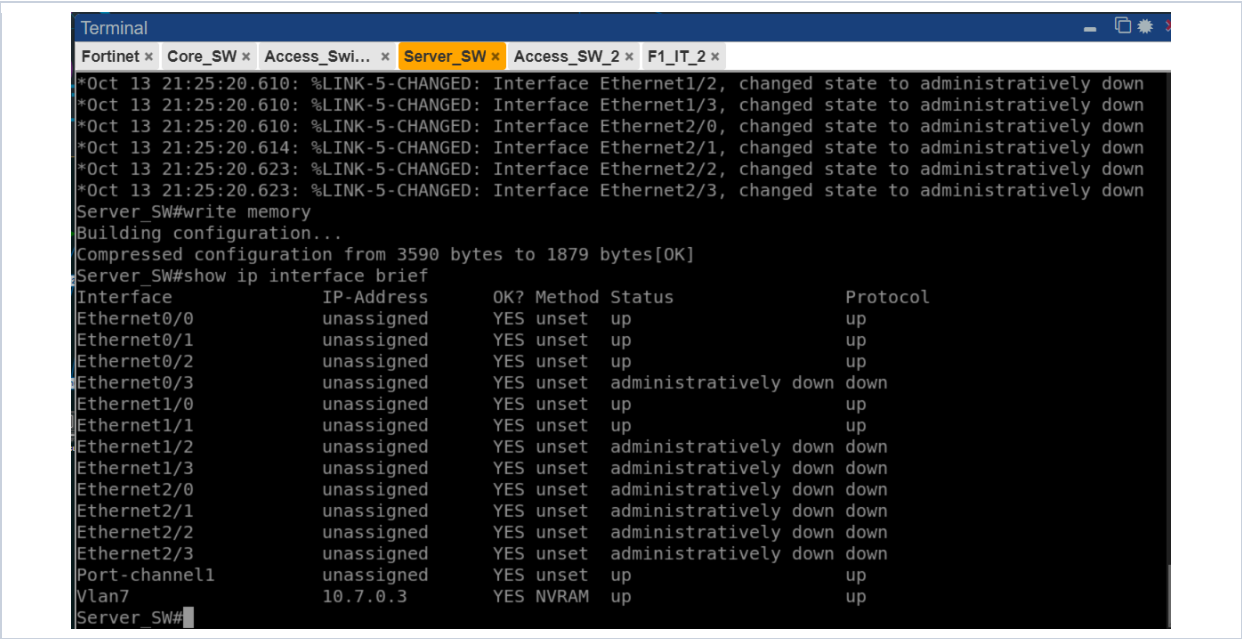


Figure 3.12: Output of `show ip interface brief` on Server_SW — shows the unused ports on Server_SW administratively down, and the Vlan7 interface with address 10.7.0.3.

3.4 Server Port Protection

Description	Apply the same protection set to server ports on Server_SW: enable BPDU Guard and Port Security with a maximum of one device per port and MAC addresses learned in sticky mode, on the Windows Server (AD) port, the Wazuh Manager port, and the Honeypot port.
Rationale	Servers are the highest-value assets in the network, and any replacement of a server or connection of a different device to its port must be detected immediately. BPDU Guard also prevents any device attached to a server port from affecting Spanning Tree stability.
Importance	Prevents impersonation of sensitive servers, ensures that the Honeypot port itself does not become a point for connecting unauthorized devices, and unifies protection standards between user ports and server ports.

Code

Cisco CLI | Server_SW

```
configure terminal
! Windows Server (Active Directory)
interface Ethernet0/0
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  exit
! Wazuh SIEM Manager
interface Ethernet1/1
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  exit
! Honeypot decoy
```

Cisco CLI | Server_SW

```
interface Ethernet1/0
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
  exit
end
write memory
```

Verification: The `show port-security` command on Server_SW lists three ports with a maximum of 1 per port and the Shutdown action on violation.

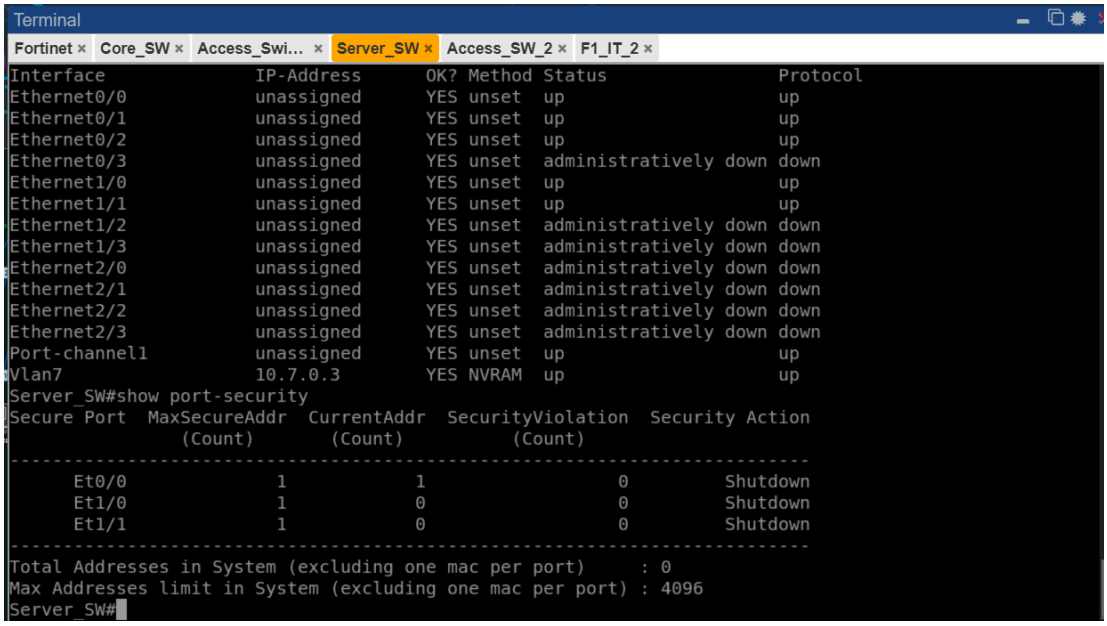


Figure 3.13: Output of `show port-security` on Server_SW — shows Port Security enabled on the three server ports with a maximum of one device per port and a Shutdown action.

Chapter 4: Administrative Device Hardening

This phase focuses on protecting the switches themselves from administrative intrusion attempts and applying the principle of least privilege. Under the Zero-Trust model, the device control plane (Management Plane) must be hardened against any unauthorized access, so a two-phase strategy was adopted:

- Phase 1 (implemented): securing local access (Local Hardening) and creating break-glass accounts.
- Phase 2 (later): integrating the switches with Active Directory to apply centralized authentication (AAA - RADIUS) once the server zone has been connected and routed.

The following settings are applied in a standardized configuration on all switches: Core_SW, Server_SW, Access_SW_1, and Access_SW_2, and the implementation

screenshots are taken from Core_SW as a model for generalizing the standards to the other devices.

4.1 Break-Glass Accounts and Password Protection

Description	Enable `service password-encryption` to hide all passwords in the running-config, and create an encrypted local administrator account with full privileges (Privilege 15), together with an encrypted password for Enable mode.
Rationale	The local account is the “last line of defense” (Break-Glass Account): it guarantees that system administrators can still log in to the switches if connectivity to the central authentication (AAA) servers is lost for any emergency reason. The `service password-encryption` command prevents passwords from appearing in clear text in configuration files, noting that it is a simple reversible obfuscation (Type 7), which is why secret is relied upon to protect the primary accounts.
Importance	Protects login credentials from leakage when configuration files are shared or viewed, and ensures that an administrative path remains available in emergencies.

Code

Cisco CLI | All Switches

```
enable
configure terminal
! Obfuscate any clear-text password in the running configuration
service password-encryption
! Local break-glass administrator account (encrypted secret)
username admin privilege 15 secret Admin@2026
enable secret Admin@2026
end
write memory
```

Verification: The `show running-config | include username` command shows that the local administrator password is stored in encrypted secret form, not in clear text.

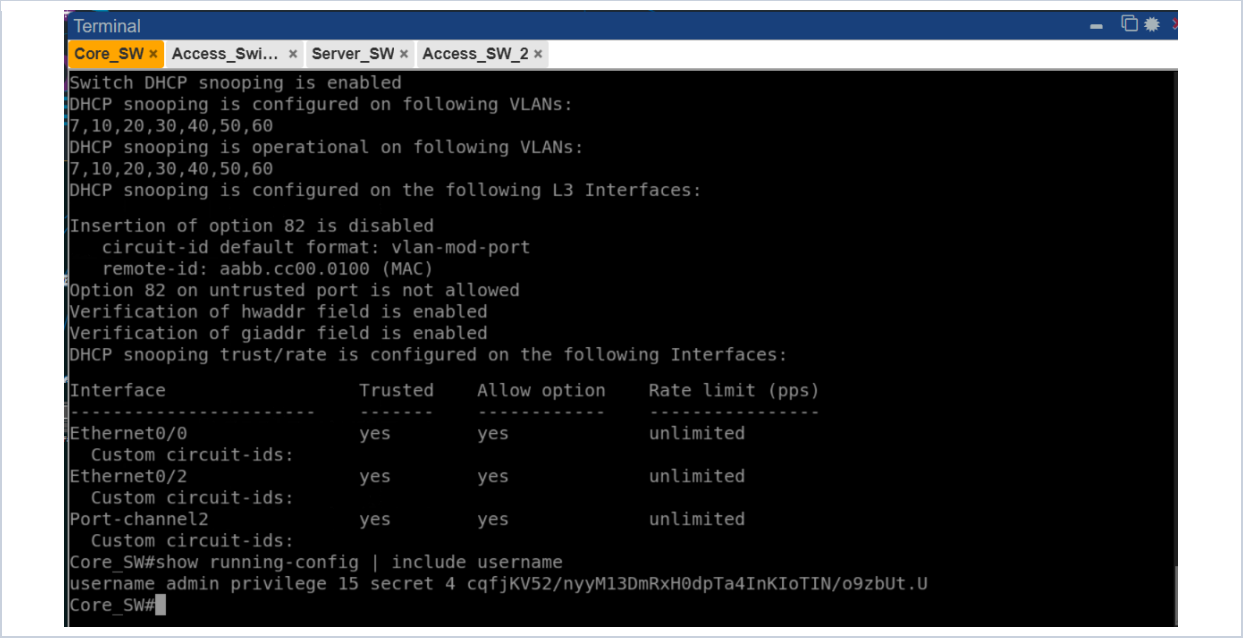


Figure 4.1: Output of `show running-config | include username` on Core_SW — shows that the local administrator password is stored as an encrypted secret and not in clear text.

4.2 Encrypting Management Sessions (SSHv2)

Description	Prepare the secure connection environment by setting the domain name AegisEud.local to match the Active Directory server's domain, generating 2048-bit RSA keys, and enabling SSH version 2.
Rationale	Aligning the domain name with the Active Directory servers prevents future FQDN name conflicts. The 2048-bit keys and SSHv2 enforce strong encryption algorithms that protect remote management sessions from decryption and tampering attacks.
Importance	Replaces the weak Telnet protocol and prevents sniffing and password theft during remote management.

Code

Cisco CLI | All Switches

```
configure terminal
! Domain name (identical to the Active Directory domain)
ip domain-name AegisEud.local
! Generate 2048-bit RSA keys
crypto key generate rsa modulus 2048
! Enable SSH version 2 (more secure)
ip ssh version 2
end
write memory
```

Verification: The `show ip ssh` command documents that SSH version 2 is enabled and displays the RSA key that was generated.

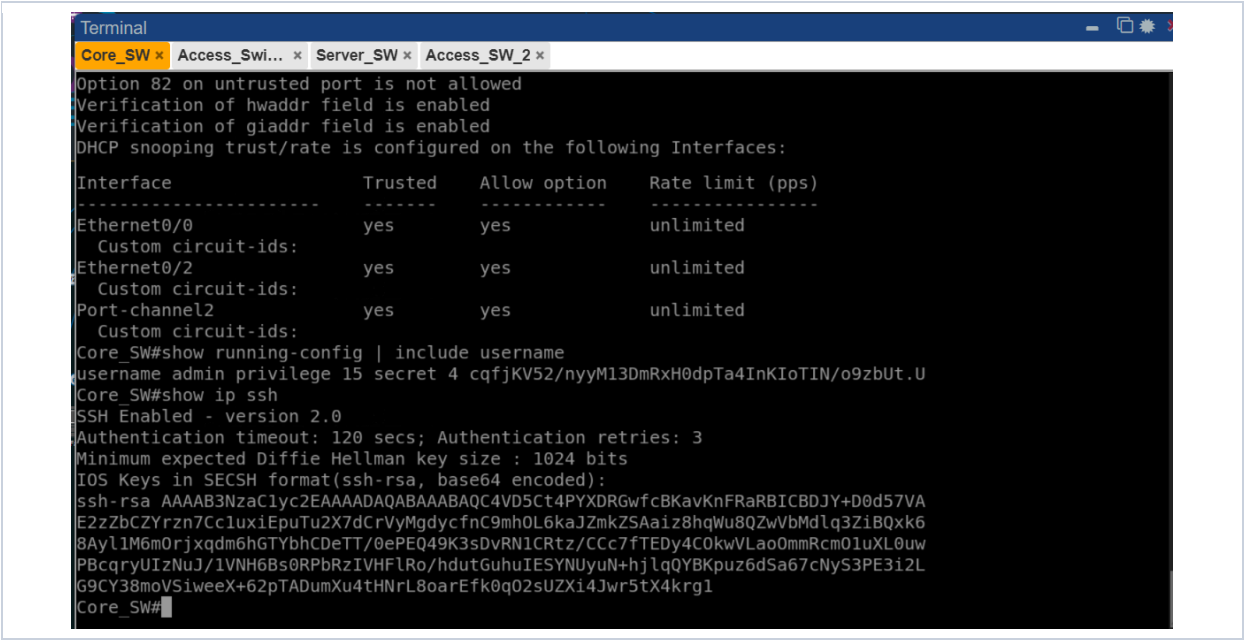


Figure 4.2: Output of `show ip ssh` on Core_SW — shows SSH version 2.0 enabled and the RSA key that was generated.

4.3 Securing Access Lines and Session Management

Description	Restrict remote access over VTY lines to the SSH protocol only with local authentication (`login local`), secure the direct Console line, enable the automatic timeout `exec-timeout 5 0` for idle sessions after 5 minutes, and enable `logging synchronous` on the Console to keep system messages from interleaving with typed commands.
Rationale	Completely close the insecure Telnet protocol to defeat sniffing attacks, and protect against abuse of unattended administrator terminals.
Importance	Reduces the chances of unauthorized access through open sessions and applies the principle of least privilege at the management level.

Code

Cisco CLI | All Switches

```
configure terminal
! Secure remote access lines (VTY) - SSH only
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
  exit
! Secure the direct console line
line console 0
  login local
  exec-timeout 5 0
  logging synchronous
  exit
end
write memory
```

Verification: The `show running-config | section line vty` command proves that remote access is restricted to the SSH protocol and that the automatic timeout is enabled.

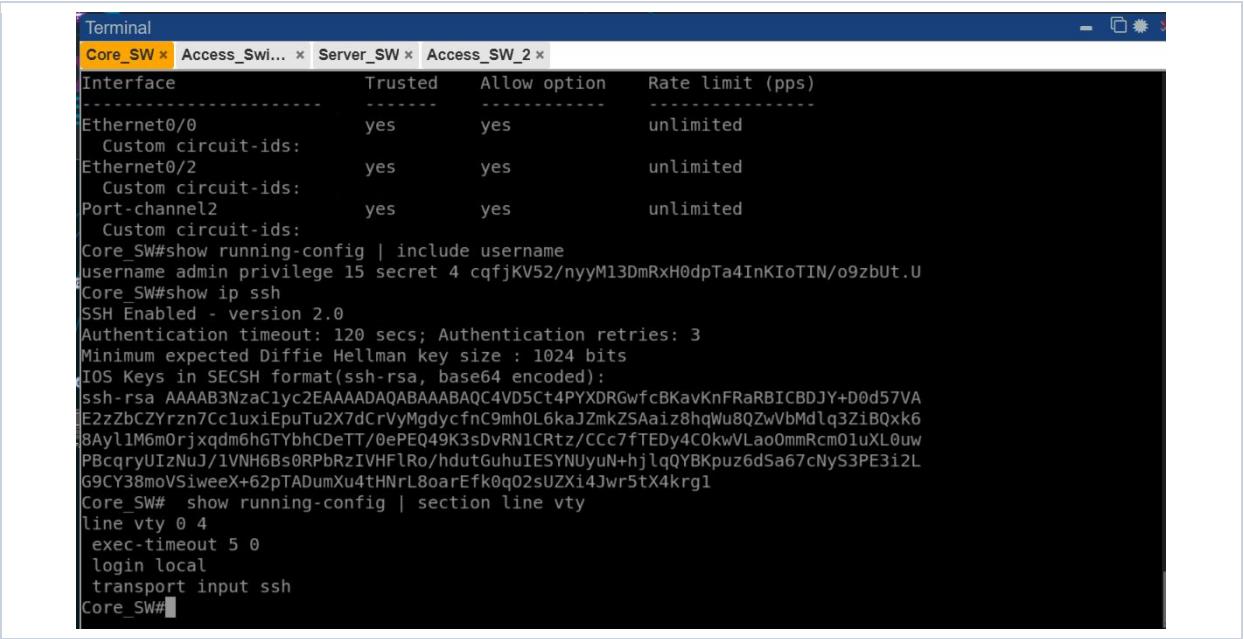


Figure 4.3: Output of `show running-config | section line vty` on Core_SW — shows that the VTY lines accept SSH only, with local authentication and a 5-minute idle timeout.

4.4 Disabling Unnecessary Services

Description	Disable the web-based management servers with the `no ip http server` and `no ip http secure-server` commands, and disable name-to-address resolution with `no ip domain-lookup`.
Rationale	Management in this design is performed exclusively over SSH, so leaving an HTTP/HTTPS server running opens an additional port for no reason. Domain lookup makes the device treat any mistyped command as a host name and try to contact a non-existent DNS server, which freezes the session and generates unintended connection attempts.
Importance	Reduces the number of open services on each device to the minimum and keeps the management session fast and stable.

Code

Cisco CLI | All Switches

```
configure terminal
! Disable web-based management (SSH only)
no ip http server
no ip http secure-server
! Do not resolve mistyped commands as host names
no ip domain-lookup
end
write memory
```

Verification: The `show running-config | include http|domain-lookup` command shows that web services and name resolution are disabled.

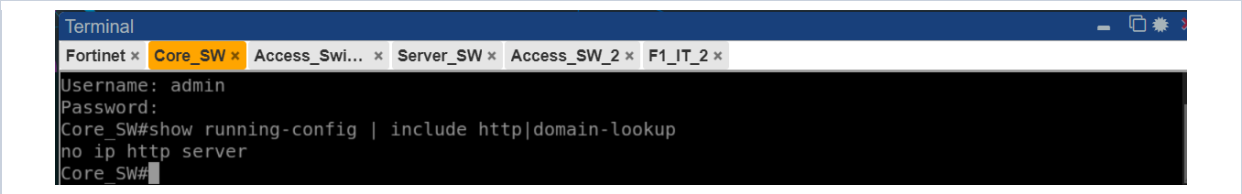


Figure 4.4: Output of `show running-config | include http|domain-lookup` on Core_SW — shows the HTTP server disabled on the device with the no ip http server command.

4.5 Login Banner

Description	Add a warning banner displayed to everyone who connects to the device, using the `banner motd` command with the text UNAUTHORIZED ACCESS PROHIBITED.
Rationale	The banner states that access is restricted to authorized personnel only and removes any claim that the user was unaware that access was unauthorized.
Importance	Supports legal accountability if a breach occurs, and is a recommended practice within Cisco device hardening standards.

Code

Cisco CLI | All Switches

```
configure terminal
! Legal warning shown to every connection attempt
banner motd # UNAUTHORIZED ACCESS PROHIBITED #
end
write memory
```

Verification: The warning banner appears when a new session is opened on the device, before login credentials are requested.

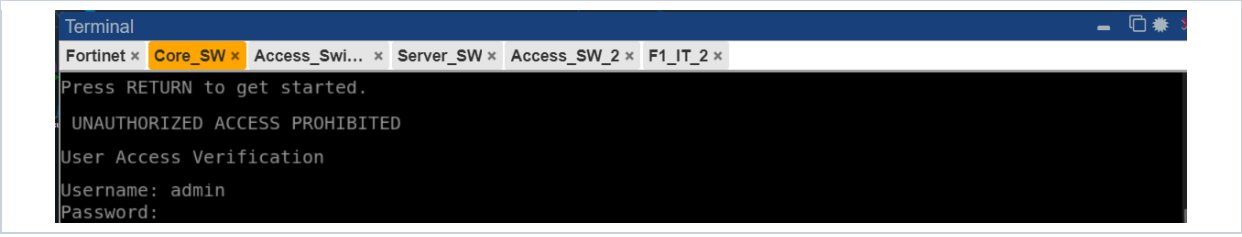


Figure 4.5: The warning banner displayed on login to Core_SW — shows the UNAUTHORIZED ACCESS PROHIBITED warning banner appearing before login credentials are requested.

4.6 Management Interface and Default Gateway (Management VLAN SVI & Default Gateway)

Description	Designate VLAN 7 (IT_SecOps) as the device management network by creating an `interface vlan 7` with a static IP address on each switch and setting the default gateway `ip default-gateway 10.7.0.1`, which is the FortiGate interface on the same network. Because the devices operate as Layer 2 switches only, `ip routing` is not enabled and no interfaces are created for any other network.
Rationale	A Layer 2 switch cannot accept an SSH connection from the network without an IP address of its own. Choosing the IT_SecOps network for management makes administrative access exclusive to the authorized IT team, and no other network can reach the devices except through an explicit FortiGate policy. A management interface also does not imply routing between networks, so all traffic between departments is still forced through the FortiGate.
Importance	Turns the SSH configuration from a service that is merely ready into actual administrative access that can be tested, without compromising the network-isolation principle on which the design is built.

Management addresses assigned to the devices:

Device	Interface	IP Address	Default Gateway
Core_SW	Vlan7	10.7.0.2/24	10.7.0.1
Server_SW	Vlan7	10.7.0.3/24	10.7.0.1
Access_SW_1	Vlan7	10.7.0.4/24	10.7.0.1
Access_SW_2	Vlan7	10.7.0.5/24	10.7.0.1

Code

Cisco CLI | Management Interface

```
! ===== Core_SW =====
configure terminal
interface vlan 7
  ip address 10.7.0.2 255.255.255.0
  exit
ip default-gateway 10.7.0.1
end
write memory
! ===== Server_SW =====
configure terminal
interface vlan 7
  ip address 10.7.0.3 255.255.255.0
  exit
ip default-gateway 10.7.0.1
end
write memory
! ===== Access_SW_1 =====
configure terminal
interface vlan 7
  ip address 10.7.0.4 255.255.255.0
  exit
ip default-gateway 10.7.0.1
end
```

Cisco CLI | Management Interface

```
write memory
! ===== Access_SW_2 =====
configure terminal
interface vlan 7
  ip address 10.7.0.5 255.255.255.0
  exit
ip default-gateway 10.7.0.1
end
write memory
```

Verification: The `show ip interface brief` command shows the Vlan7 interface with its address and an up/up state, `ping 10.7.0.1` proves connectivity to the gateway, and a PuTTY SSH connection from the device F1_IT_2 inside VLAN 7 to address 10.7.0.2 proves a successful login to the device.

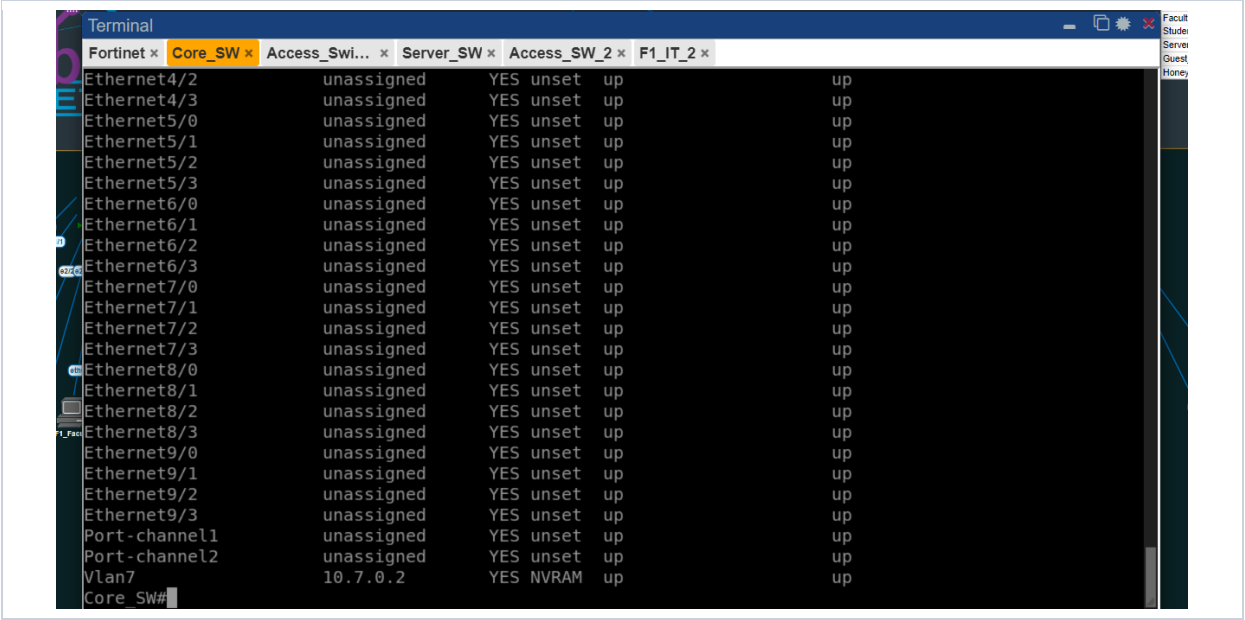


Figure 4.6: Output of `show ip interface brief` — shows the Vlan7 interface on Core_SW with address 10.7.0.2 and an up/up state.

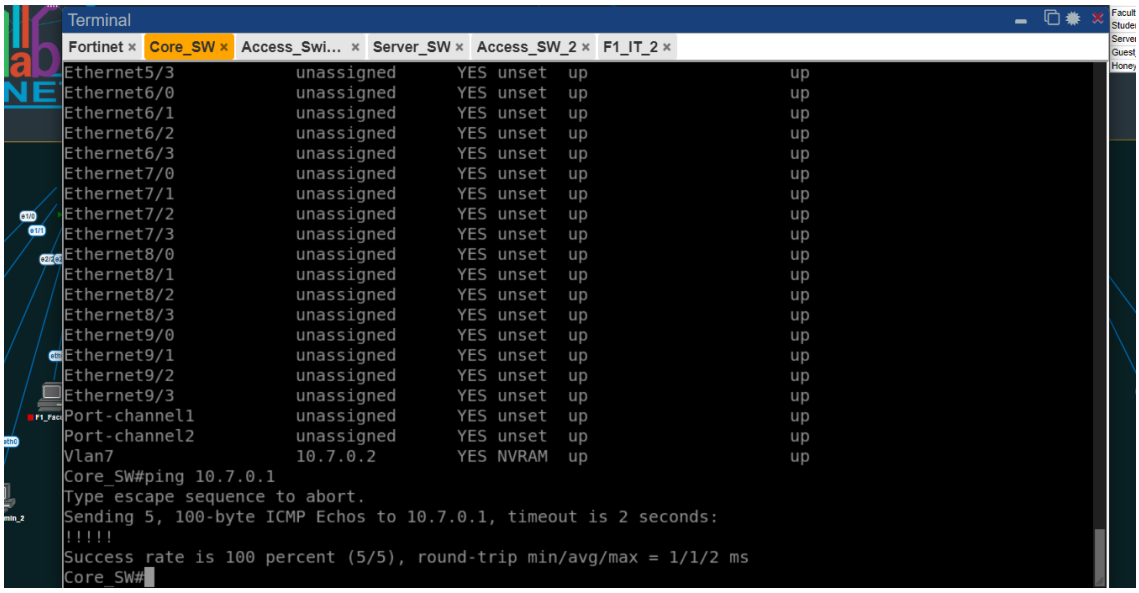
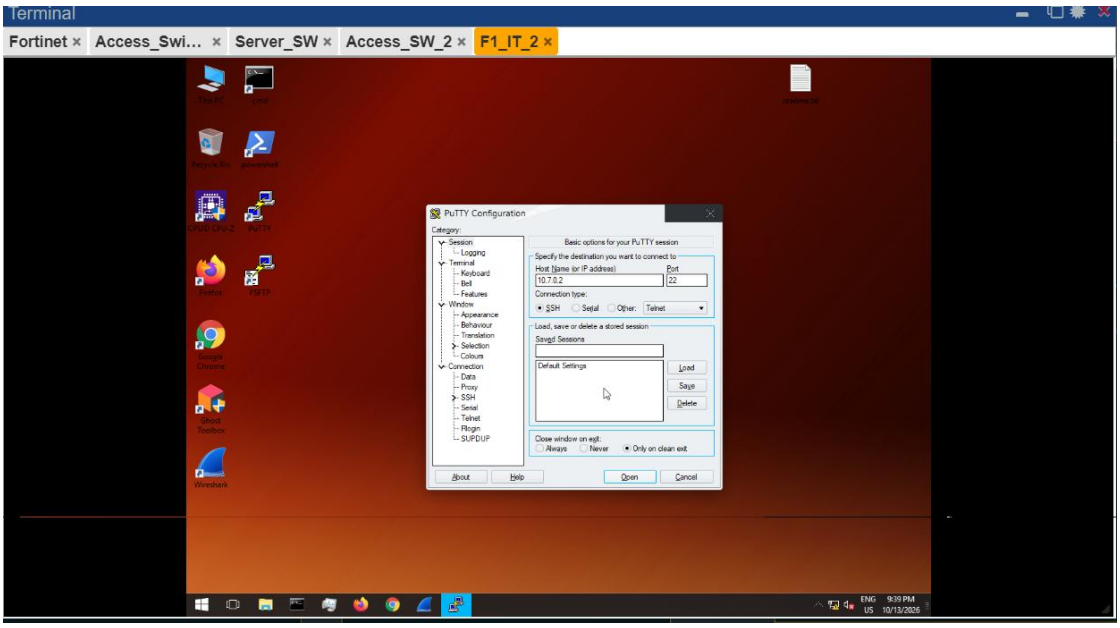


Figure 4.7: Successful reachability of the gateway 10.7.0.1 from Core_SW using `ping` — shows successful connectivity to the gateway 10.7.0.1 with a 100 percent success rate (5 of 5).



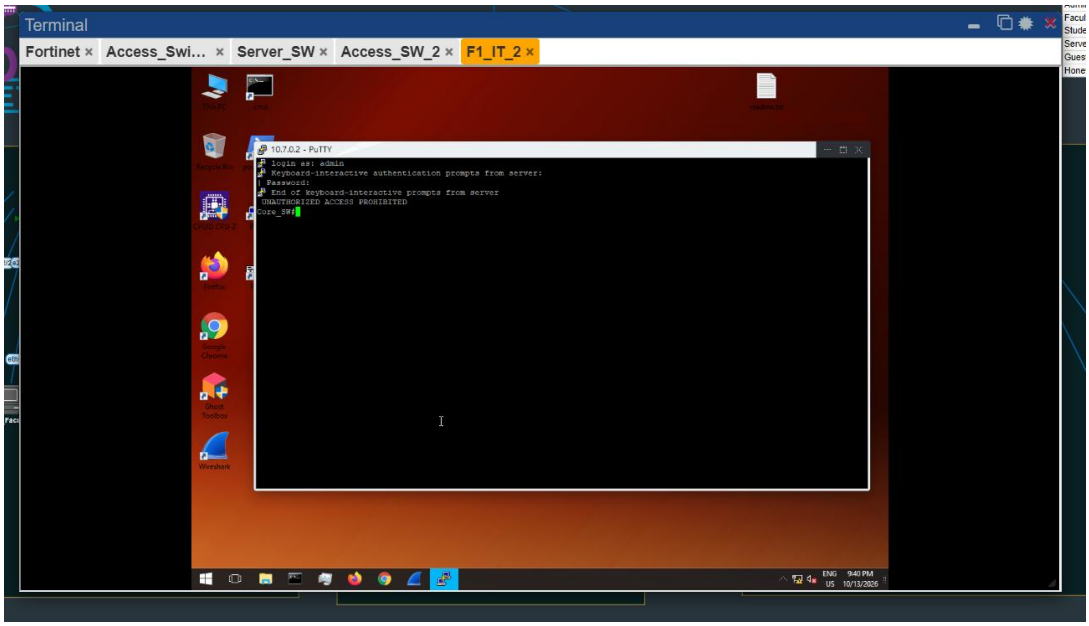


Figure 4.8: SSH session using PuTTY from device F1_IT_2 inside VLAN 7 to Core_SW (10.7.0.2) — shows the PuTTY connection setup to 10.7.0.2, then the successful SSH session with the warning banner.

4.7 Centralized Authentication (AAA - RADIUS)

Status	The second phase of the management-security strategy, to be implemented once the server zone has been connected and routed.
Description	Integrate the switches with Active Directory servers over the RADIUS protocol to restrict login and network-configuration privileges to the authorized IT team only, while the local Break-Glass account remains as a fallback path.
Rationale	Move identity and privilege management to a central point instead of local accounts distributed across every device.
Importance	Provides a central record of who logged in to the devices and what they changed, and makes it easy to revoke privileges immediately when needed.

Chapter 5: Routing and Architectural Integration (Layer 3 & FortiGate Integration)

This phase prepares the infrastructure for integration with advanced security systems and the artificial intelligence teams. In a Zero-Trust architecture, no internal data may be routed between networks (inter-VLAN routing) without passing through security checkpoints; therefore Layer 3 capabilities were removed from Core_SW and assigned entirely to the FortiGate, making it the central router and the security controller of data flow paths and DHCP address distribution.

5.1 Blocking Internal Routing (Disabling Inter-VLAN Routing)

Description	A strict architectural decision to disable Layer 3 routing capabilities on Core_SW so that it operates as a Layer 2 switch only, with all network gateways located on the FortiGate.
Rationale	Force all data exchanged between departments (East-West Traffic) to pass through the FortiGate; no department talks to another until the data has undergone deep packet inspection and security policies have been applied.
Importance	Prevents any bypass path between departments that escapes monitoring, which is the essence of applying Zero-Trust inside the network. This decision is complemented by the DHCP Snooping system described in Section 3.2.

5.2 Initial Firewall Provisioning (Initial Provisioning & Hardware Tuning)

Description	Perform the initial login to the FortiGate and change the default password to access the graphical interface (GUI), then assign a static IP address to the management interface (Port 1). The type of the FortiGate's virtual network cards (QEMU NIC) was also changed to e1000.
Rationale	A software conflict was discovered in PNETLab between the speed of the FortiGate's virtual network cards intended for virtual environments (10Gbps Virtio) and the speed of the Cisco IOL switches (1Gbps), which obstructs link-aggregation negotiation. The constraint was resolved by forcing a uniform speed of 1Gbps between all core devices.
Importance	Ensures stable connectivity between the FortiGate and Core_SW, and makes the behavior of the simulation environment match that of real environments.

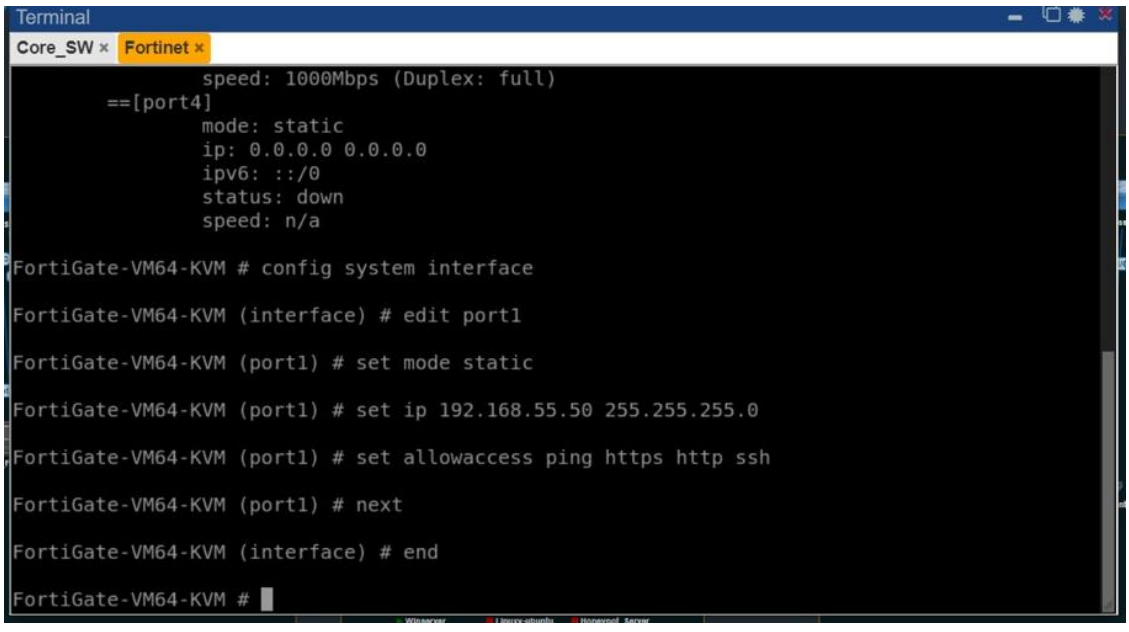
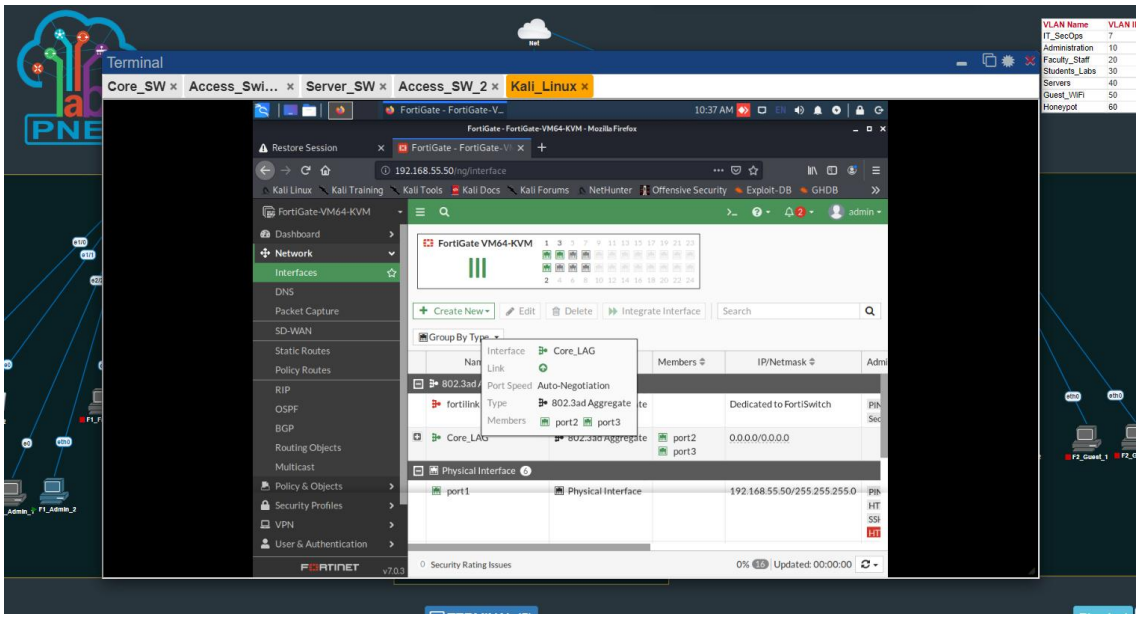


Figure 5.1: Static IP address configuration for the management interface Port 1 on the FortiGate — shows a static IP address assigned to the management interface port1 and the access methods allowed on it.

5.3 Establishing the Aggregated Transit Path (Static Link Aggregation - Core_LAG)

Description	The following steps were performed from the FortiGate graphical interface: 1) Navigate to Network > Interfaces. 2) Create an Aggregate Interface named Core_LAG. 3) Assign ports port2 and port3 as members of this interface. 4) Set the aggregation mode to Static / Mode ON to match the Core_SW configuration.
Rationale	Guarantee high availability and widen the bandwidth between Core_SW and the FortiGate to avoid a bottleneck. In real environments LACP is preferred, but a software bug in the simulator drops LACP negotiation packets (LACP PDUs), so aggregation was set to Static / Mode ON on both ends.
Importance	This resulted in the channel being built successfully and in a stable data path at 2Gbps, unaffected by the simulator's limitations.



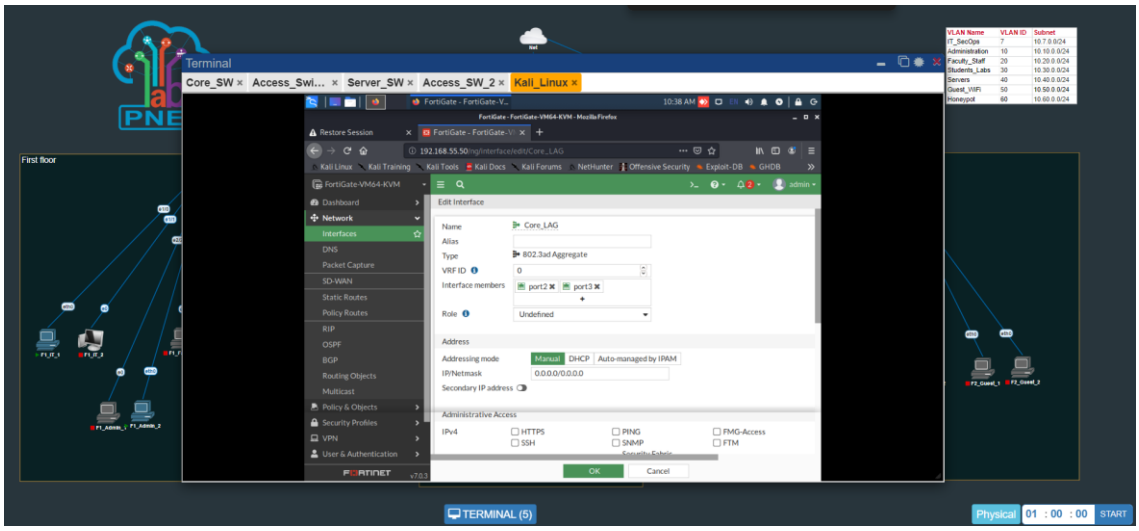


Figure 5.2: Creating the aggregate interface Core_LAG with members port2 and port3 on the FortiGate — shows the creation of the aggregate interface Core_LAG of type 802.3ad Aggregate with members port2 and port3.

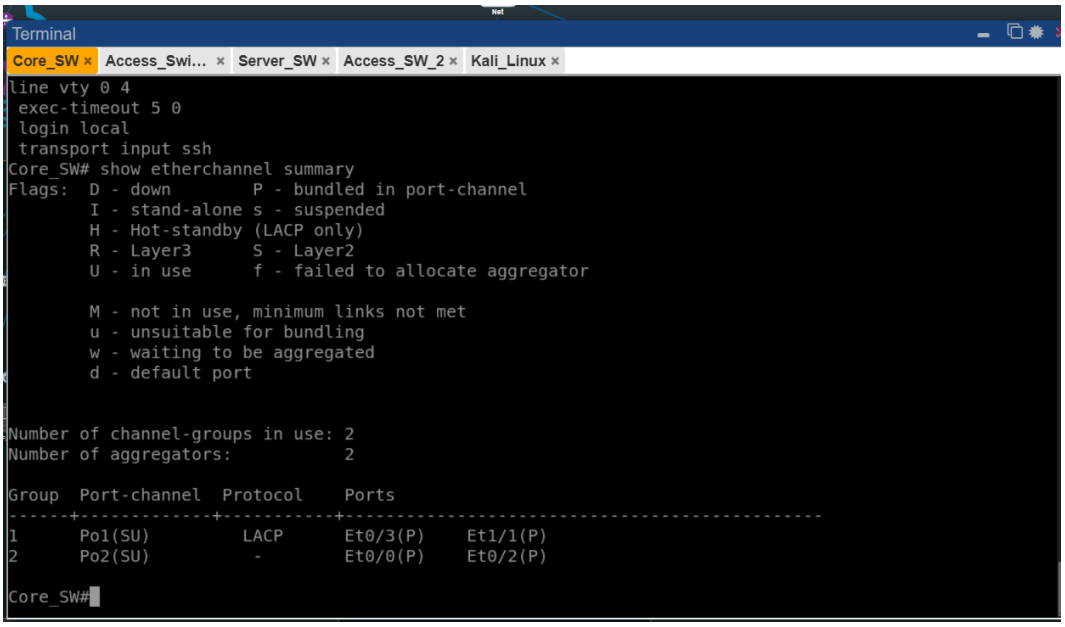


Figure 5.3: Output of `show etherchannel summary` on Core_SW — shows Port-channel 2 in state SU with no protocol (static aggregation) and members Et0/0 and Et0/2.

5.4 Logical Addressing Scheme (VLAN-to-Subnet Mapping Scheme)

Description	The infrastructure team adopted an engineering strategy for address allocation instead of the traditional device-count-based subnetting (VLSM): Range structure: the 10.0.0.0/8 range was reserved for the project and a /24 mask was assigned to each department, so that the second octet of the IP address matches the department's VLAN ID. Example: the Administration network (VLAN 10) takes the range 10.10.0.0/24. Default gateway and distribution: the first available address (.1) was chosen as the FortiGate's Default Gateway for all networks, the automatic DHCP distribution range was set from .2 to .254, which is the range applied on the FortiGate as shown in Figure 5.4.
Rationale	Ease of management, future scalability, and path security, since the department can be identified directly from the IP address.
Importance	Makes it easier to trace data paths and to apply Firewall policies between departments with great precision.

Architectural distribution table (IP Addressing & DHCP Scheme):

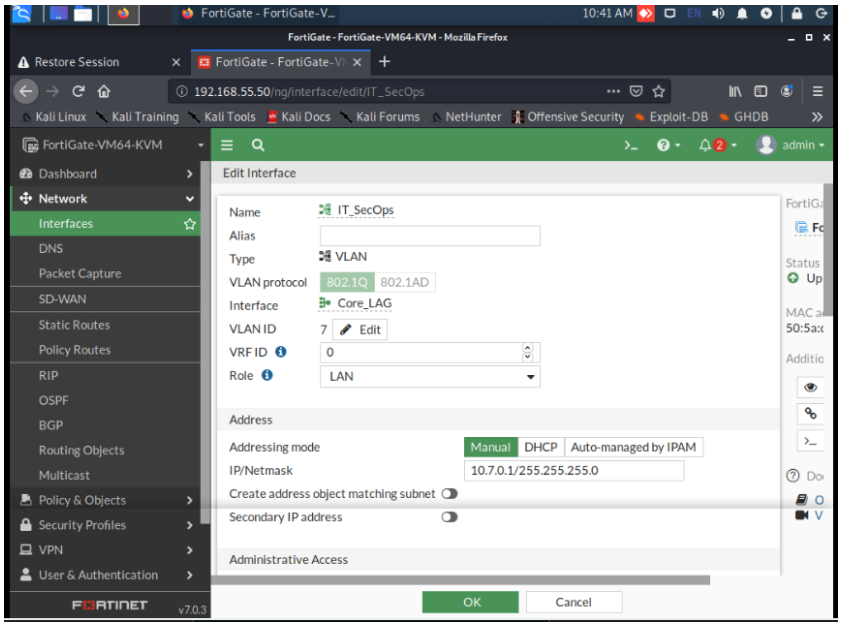
VLAN Name	VLAN ID	Gateway	DHCP Range
IT_SecOps	7	10.7.0.1/24	10.7.0.2 - 10.7.0.254
Administration	10	10.10.0.1/24	10.10.0.2 - 10.10.0.254
Faculty_Staff	20	10.20.0.1/24	10.20.0.2 - 10.20.0.254
Students_Labs	30	10.30.0.1/24	10.30.0.2 - 10.30.0.254
Servers	40	10.40.0.1/24	10.40.0.2 - 10.40.0.254
Guest_WiFi	50	10.50.0.1/24	10.50.0.2 - 10.50.0.254
Honeypot	60	10.60.0.1/24	10.60.0.2 - 10.60.0.254

5.5 Configuring Sub-interfaces and DHCP Servers (VLAN Sub-interfaces & DHCP)

Description	Based on the architectural table, the aggregate interface Core_LAG was divided into sub-interfaces, each serving a department in complete isolation. The following general steps were applied to create the interfaces for the seven VLANs: 1) Go to Network > Interfaces, then click Create New > Interface. 2) Set the interface Type to VLAN and bind it to the parent interface, Core_LAG. 3) Assign the VLAN ID matching the department and enter an IP/Netmask to serve as the department's default gateway. 4) Enable the DHCP Server so that it distributes addresses automatically according to the range defined for each network. 5) Restrict Administrative Access on all interfaces by allowing only the PING protocol for connectivity testing, except for the IT_SecOps network (VLAN 7), which was granted HTTPS, SSH, and PING privileges.
Rationale	Apply the principle of least privilege so that control of the FortiGate is available only to the authorized administration team, while still allowing connectivity to be tested from the other departments.
Importance	Makes the FortiGate the single gateway for all departments, which allows deep packet inspection policies to be applied between all networks.

Comprehensive worked example: creating the IT_SecOps network (VLAN 7) as a model of the preceding steps:

Setting	Value
Name	IT_SecOps
Interface	Core_LAG
VLAN ID	7
IP/Netmask	10.7.0.1/24
Administrative Access	PING , HTTPS , SSH
DHCP Server	Enabled (10.7.0.2 - 10.7.0.254)



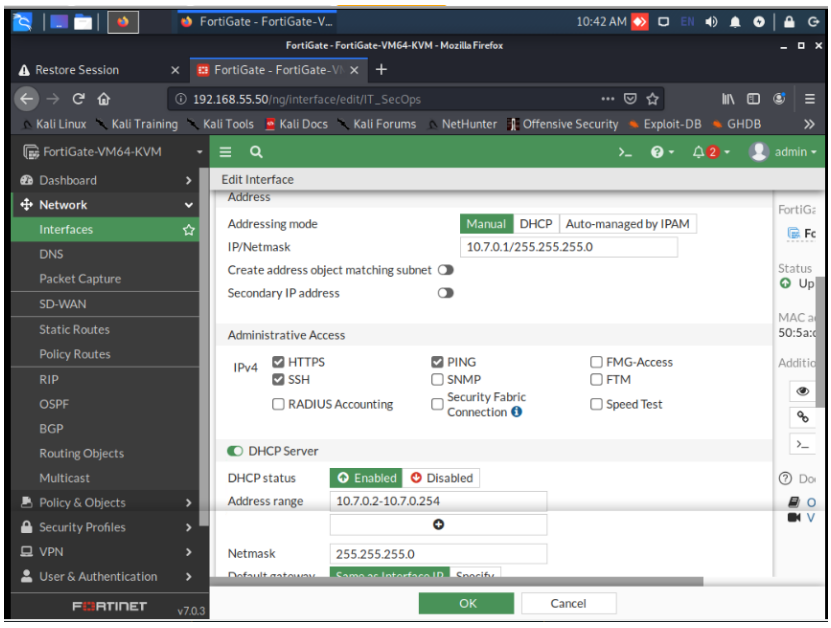


Figure 5.4: Creating the VLAN 7 (IT_SecOps) interface — shows the IT_SecOps interface settings: VLAN ID 7, address 10.7.0.1/24, HTTPS, SSH, and PING access, and the DHCP Server enabled.

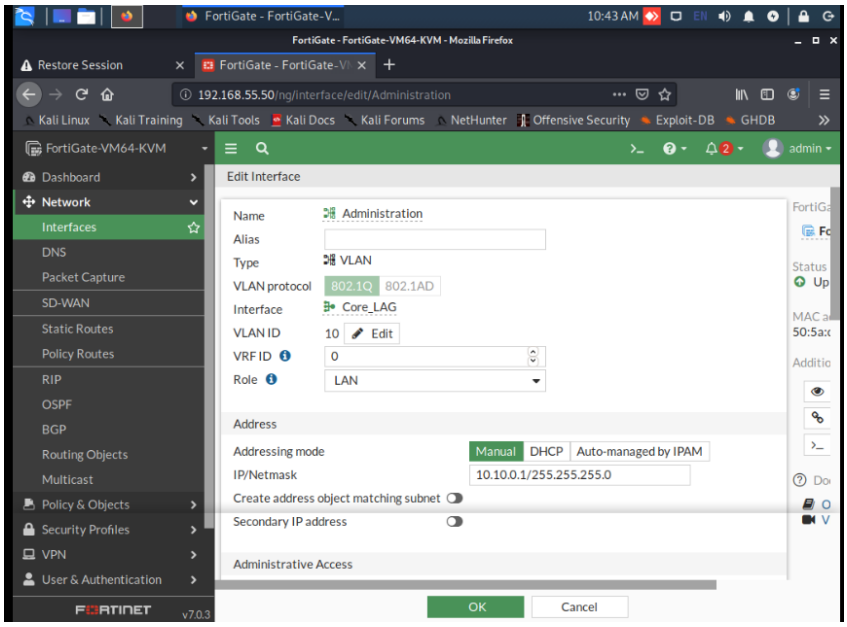
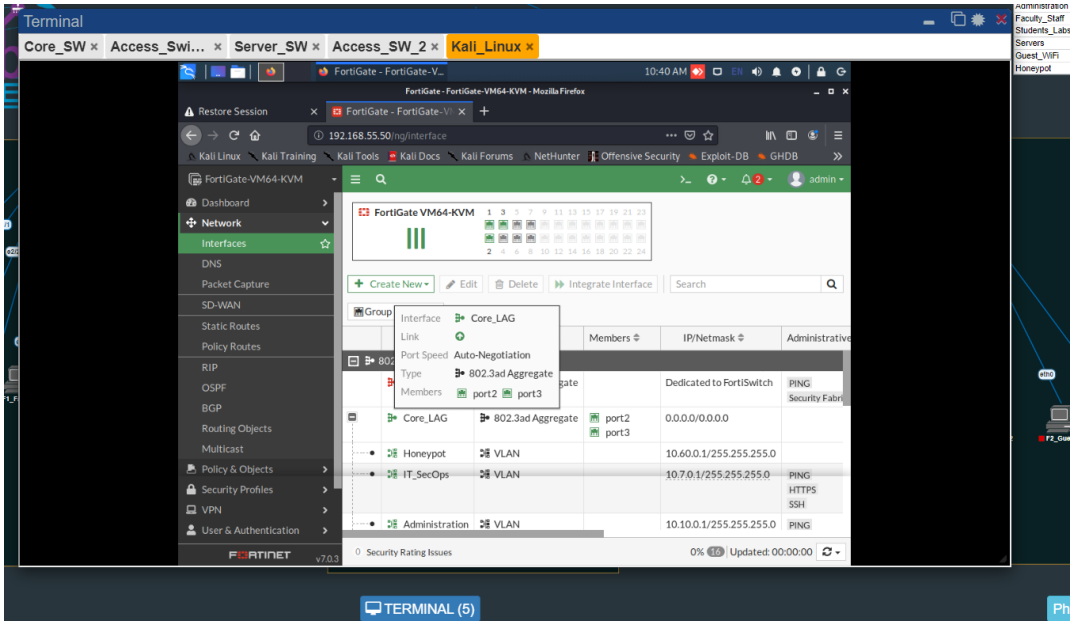


Figure 5.5: Creating the Administration (VLAN 10) interface using the same steps for confirmation and comparison — shows the creation of the Administration interface with VLAN ID 10 and address 10.10.0.1/24 using the same steps.



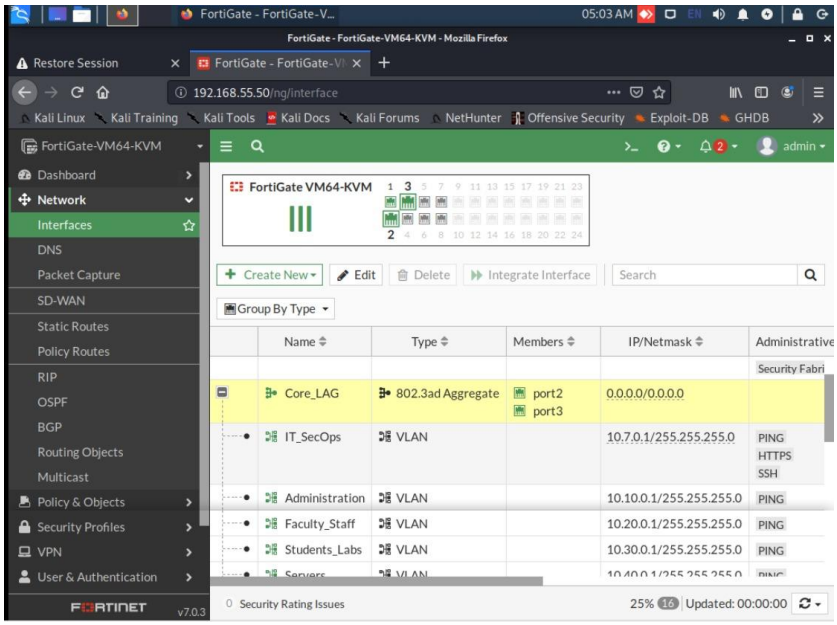
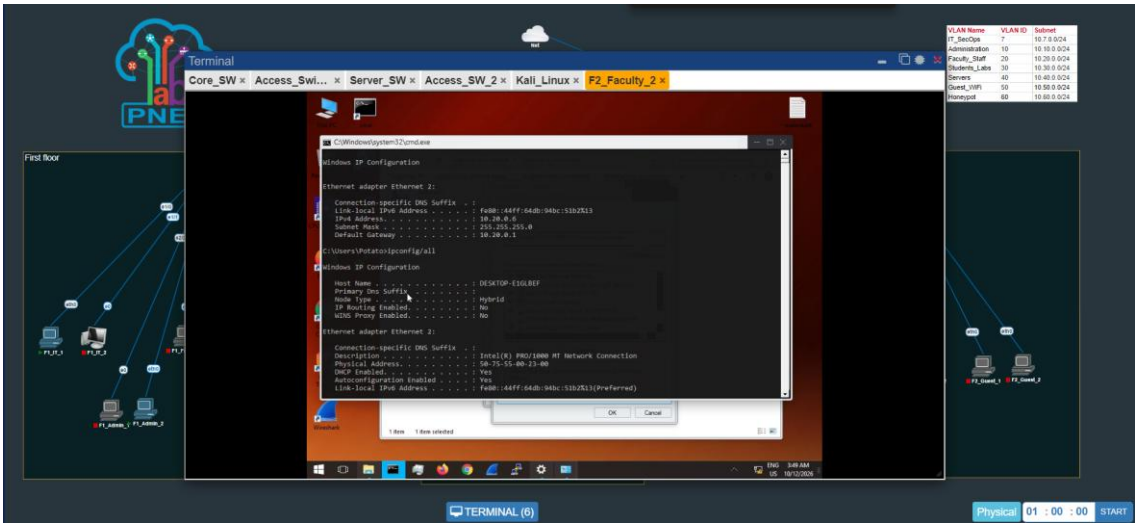


Figure 5.6: The Interfaces page on the FortiGate — shows the Core_LAG interface and the department networks branching from it, with their gateway addresses and the administrative access allowed on each interface.

5.6 End-to-End Connectivity Test (End-to-End DHCP Verification)

Description	A workstation PC named F2_Faculty_2 in the faculty department on the second floor (VLAN 20) was started and a DHCP Discover request was sent. The device successfully received an IP address within the department's correct range, 10.20.0.6, and received the FortiGate's default gateway address, 10.20.0.1.
Rationale	Verify the integrity of the entire architecture, passing through Core_SW and the Access switches secured with DHCP Snooping, up to the FortiGate successfully performing the role of central router and DHCP server.
Importance	Proves in practice that the entire protection chain works together without blocking legitimate connectivity.



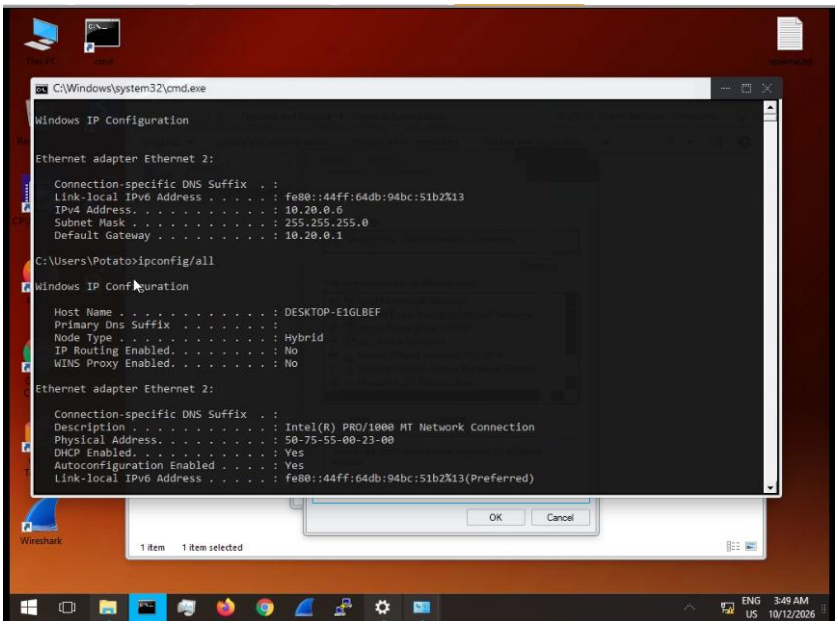


Figure 5.7: Output of `ipconfig /all` on device F2_Faculty_2 (VLAN 20) — shows the device receiving address 10.20.0.6 and gateway 10.20.0.1 via DHCP from the FortiGate.

Chapter 6: Strategic Architecture Decisions

This chapter highlights the design decisions that were taken deliberately to serve the project concept and to apply the Zero-Trust Architecture model; these decisions are entirely independent of the constraints of the implementation environment.

6.1 Centralizing the DHCP Server on the FortiGate Instead of Active Directory

Engineering Decision	The DHCP address-distribution server was placed on the FortiGate instead of the Active Directory server in the server zone.
Security Rationale	In the zero-trust model, transit gateways must be secured early. If DHCP were on the AD, unknown devices would be allowed to reach the sensitive server zone (VLAN 40) merely to obtain an IP address, exposing the AD to DHCP Starvation attacks.
Importance	Moving the service to the FortiGate makes it the absolute “gatekeeper” that distributes addresses and applies blocking policies immediately, isolating unknown devices from the critical infrastructure.

6.2 Moving Trust to the Firewall (DHCP Snooping Core-Trust)

Engineering Decision	DHCP Snooping inspection was enabled centrally, designating the aggregate interface leading to the FortiGate as the “only trusted” path (Trusted Port) (see Section 3.2).
Security Rationale	Prevent any attempt to plant a rogue server inside the internal network, and ensure that the FortiGate, the security gateway, is the only legitimate distributor of IP addresses.
Importance	Strengthens central control and prevents address spoofing attacks.

6.3 Isolating the Decoy Server in a Dedicated Zone (Honeypot Isolation)

The Honeypot is a decoy device deliberately deployed inside the network with attractive fake services and intentional weaknesses, and with no real operational function. Since no legitimate user ever needs to connect to it, any connection that reaches it is suspicious by definition.

Engineering Decision	The decoy server was placed in its own network, VLAN 60 (Honeypot), with gateway address 10.60.0.1, connected to port Ethernet1/0 on Server_SW and kept apart from the real server network, VLAN 40.
Rationale	Complete separation between the real assets and the decoy: if the honeypot is compromised, the attacker finds no direct path to Active Directory or Wazuh Manager. A dedicated network also allows specific FortiGate policies that block any traffic leaving the honeypot zone toward other networks (Containment).
Importance	Provides a high-fidelity early warning with the lowest false-positive rate, reveals the reconnaissance and scanning stage that traditional controls struggle to detect, allows the attacker's behavior and tools to be studied away from the real systems, and feeds Wazuh SIEM and the AI models with realistic attack data.

Network Controls Applied to the Honeypot Zone

- An independent network, VLAN 60, that crosses only the necessary trunk links (Port-channel 1 and Port-channel 2) as described in Section 2.8.
- Securing port Ethernet1/0 with Port Security and BPDU Guard as described in Section 3.4.
- Enabling DHCP Snooping for VLAN 60 on Server_SW and Core_SW as described in Section 3.2.
- A dedicated sub-interface on the FortiGate with address 10.60.0.1/24 and PING-only administrative access, as described in Section 5.5.

Containment is completed at Layer 3 by FortiGate policies that prevent the honeypot from reaching any internal network or the Internet; these fall within the firewall team's scope. The honeypot must never contain any real data or passwords.

Events generated by the honeypot and detected by Wazuh:

Indicator	Source	Meaning
Inbound connection to VLAN 60	FortiGate / Honeypot	Reconnaissance activity or a possible attack inside the lab.
Download of a booby-trapped file (Honeytoken)	Canarytokens	Human or automated interaction with a file that looks sensitive.
Outbound connection attempt from the honeypot	FortiGate	An attempt to use the honeypot as a pivot toward other networks.

Chapter 7: Simulation-Environment Engineering Constraints and Real-World Alternatives (Constraints & Mitigations)

Because the project relies on the PNETLab virtual simulation environment with Cisco IOL / vIOS images, the infrastructure team encountered several technical constraints related to hardware (Hardware ASICs). Engineering alternatives were adopted for these constraints, while stressing the necessity of the original technologies in real-world environments.

7.1 Dynamic ARP Inspection and IP Source Guard

Constraint and Alternative	The virtual switch images do not support these technologies stably. As an alternative, strict reliance was placed on Port Security (Max 1, Sticky) to prevent fake MAC address generation tools and to shut the port down immediately.
Necessity in a Real Environment	They must be enabled to provide proactive protection that stops ARP Poisoning and IP Spoofing attacks directly at the user port, before they reach the network.

7.2 Wireless Controllers and Access Points (Wireless Controller & Access Points)

Constraint and Alternative	The PNETLab environment is not designed to simulate radio waves (RF) accurately. As an alternative, the guest network was designed as an isolated wired zone (VLAN 50).
Necessity in a Real Environment	A WLC must be used with 802.1x authentication enabled and tied to a RADIUS / AD server to ensure that no one can break into the wireless network, and to detect Rogue Access Points.

7.3 Dynamic Link-Aggregation Protocols (LACP 802.3ad)

Constraint and Alternative	A software bug was observed in the simulation environment that causes the LACP negotiation packets (LACP PDUs) exchanged between Cisco IOL and the QEMU FortiGate to be dropped. As an engineering alternative, the aggregation mode between them was changed to static aggregation (Static EtherChannel / Mode ON), and the interface was renamed Core_LAG to ensure successful connectivity and stable data transfer.
Necessity in a Real Environment	LACP should always be preferred for link aggregation, because it can intelligently detect unidirectional link faults and automatically isolate faulty ports to prevent data loss.

7.4 Interface Speed Mismatch

Constraint and Alternative	A mismatch between the speed of the FortiGate's virtual network cards (10Gbps Virtio) and the speed of the Cisco IOL switches (1Gbps) obstructs the success of link-aggregation negotiation. As an alternative, the QEMU NIC type was changed to e1000 to unify the speed at 1Gbps.
Necessity in a Real Environment	The speeds of facing interfaces must be unified before setting up aggregation to ensure successful negotiation and a stable channel.

Chapter 8: Conclusion

8.1 Summary of Achievements

This report provided complete documentation of building and securing the network infrastructure according to the Zero-Trust Architecture model, through which the following was achieved:

- Building a stable, highly available foundation layer using Rapid-PVST+, EtherChannel, and Trunk 802.1Q with Native VLAN 444.
- Segmenting the network into 7 isolated security zones that limit lateral movement.
- Securing user ports with Port Security, BPDU Guard, DHCP Snooping, and DTP Nonegotiate, and shutting down unused ports.
- Hardening the Management Plane with SSHv2, break-glass accounts, password obfuscation, and automatic session timeouts, along with disabling unnecessary services and a legal warning banner.
- Restricting the allowed VLANs on each trunk and disabling VLAN propagation with VTP Transparent mode.
- Designating VLAN 7 as the management network with SVI interfaces and static addresses, without enabling any routing on the switches.

- Shutting down unused ports and moving them to VLAN 444, and securing server ports with Port Security and BPDU Guard.
- Isolating the Honeypot in VLAN 60 with port protection and DHCP Snooping so that it acts as a high-fidelity early-warning source.
- Delegating inter-network routing and DHCP distribution to the FortiGate so that every exchange of data between departments is subject to security inspection.
- Documenting the simulation constraints and the engineering alternatives, and what corresponds to them in real-world environments.

8.2 Next Steps

- Integrating the switches with Active Directory to apply centralized AAA - RADIUS authentication.
- Applying FortiGate deep packet inspection policies between departments.
- Integrating the infrastructure with Wazuh SIEM and AI models for monitoring and analysis.
- Restricting administrative access with an `access-class` access list on the VTY lines, and using the scrypt algorithm to protect secrets.
- Enabling Dynamic ARP Inspection, IP Source Guard, WLC with 802.1x, and LACP when moving to a real working environment.

8.3 Final Remarks

In this way, a solid architectural foundation was laid, built on isolation and continuous verification instead of implicit trust; every port, path, and device is subject to a documented security policy with its rationale and importance, and the network is ready for the next phase of intelligent threat monitoring and analysis.